# 1.40 Trigonometry: $\sin x$, $\cos x$, $\tan x$ in Degrees

Worked Solutions

**Notes on the method used here for solving trigonometric equations:**

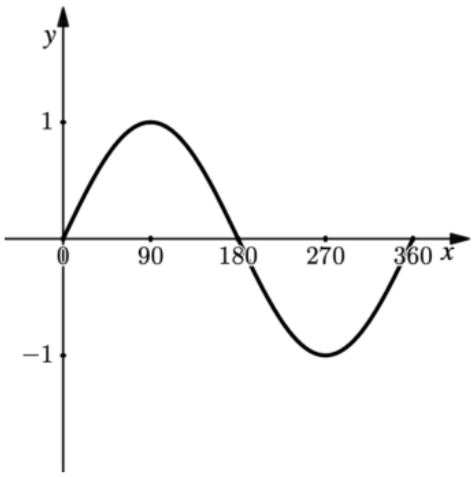
The method used here may in some instances appear like a long-winded way of getting to the required solutions, but this is to construct a general approach that will work for all situations, including more complicated examples.

For sine and cosine equations, we start by finding the two solutions in the "first period" of the function. For example, for $\sin x = 0.3$ this would be the two solutions within $0° \leq x < 360°$, and for $\cos 2x = 0.6$ this would be the two solutions within $0° \leq x < 180°$. Then we add and/or subtract the function's period to find all the required solutions.

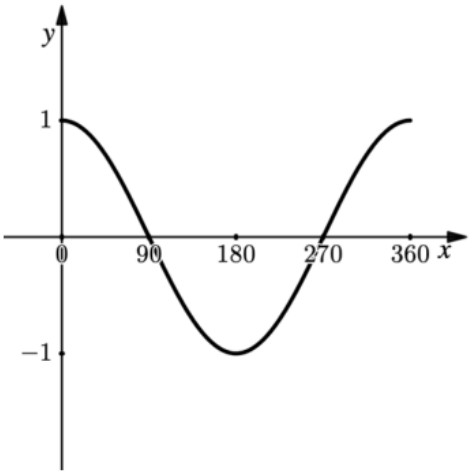
For tangent equations, we start by finding the first solution. For example, for $\tan x = 2$ this would be $x = \arctan 2$, and for $\tan 3x = 6.4$ this would be $x = \frac{1}{3} \arctan 6.4$. Then we add and/or subtract the function's period to find all the required solutions.

The diagrams used in the questions are for illustrative purposes only to help spot symmetries and the horizontal lines do not always reflect the scaling in the question.

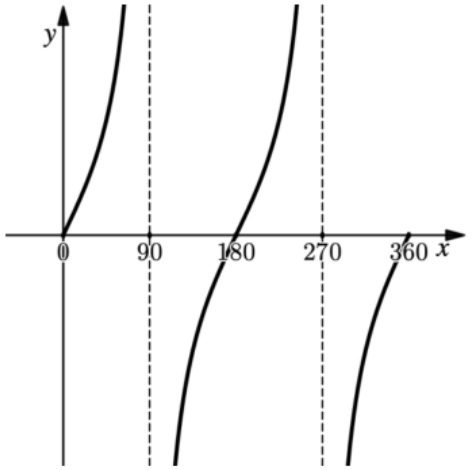
**1 a)**



**1 b)**

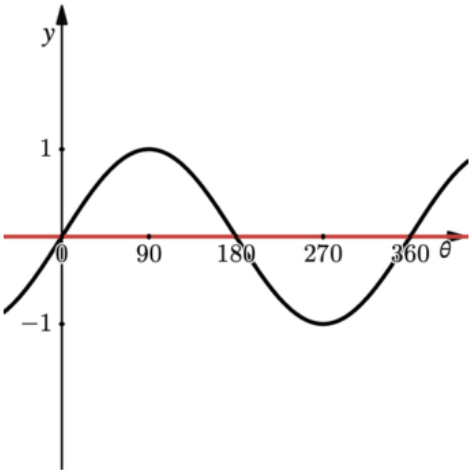



**1 c)**



**2 a)**

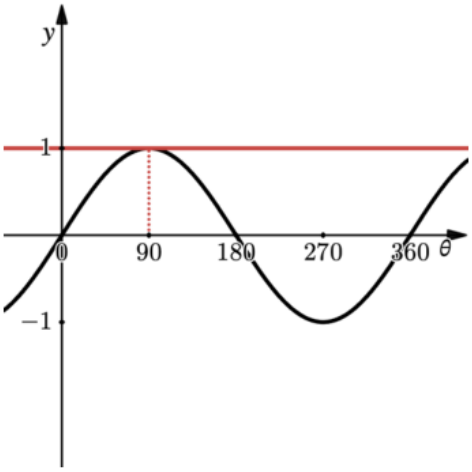
$\sin\theta = 0$ for $0° \leq \theta < 360°$.




$\Rightarrow \theta = \{0°, 180°\}$.

**2 b)**

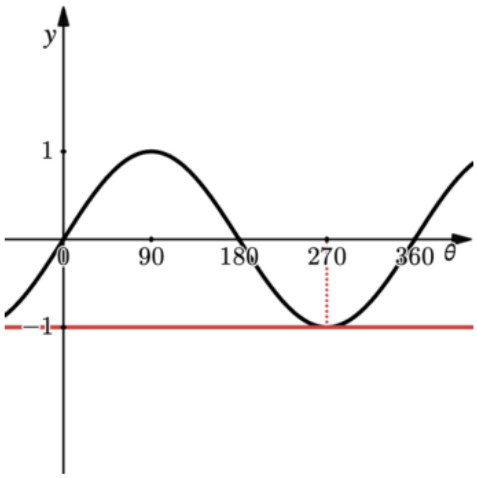
$\sin\theta = 1$ for $0° \leq \theta < 360°$.




$\Rightarrow \theta = \{90°\}$.

**2 c)**

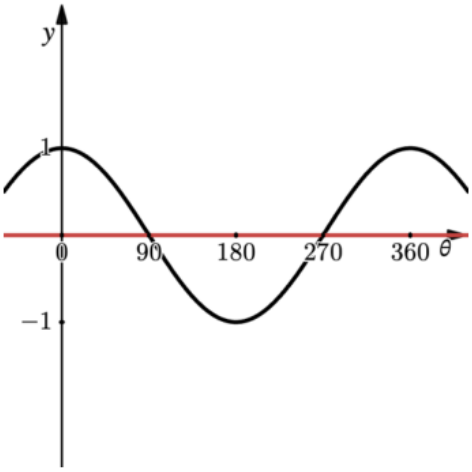
$\sin\theta = -1$ for $0° \leq \theta < 360°$.




$\Rightarrow \theta = \{270°\}$.

**3 a)**

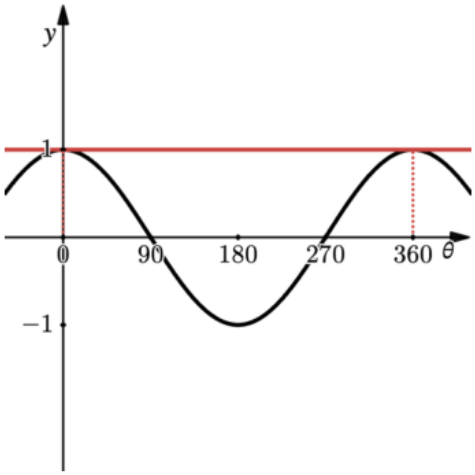
$\cos\theta = 0$ for $0° \leq \theta < 360°$.




$\Rightarrow \theta = \{90°, 270°\}$.

**3 b)**

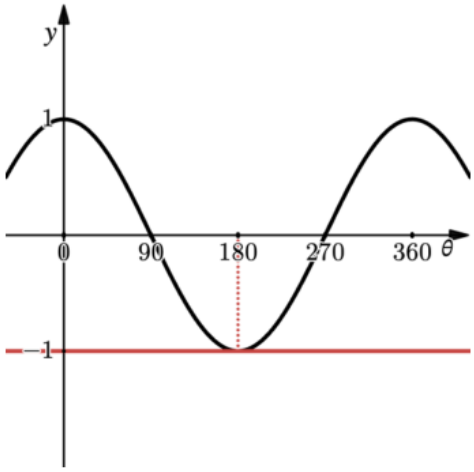
$\cos\theta = 1$ for $0° \leq \theta < 360°$.




$\Rightarrow \theta = \{0°\}$.

**3 c)**

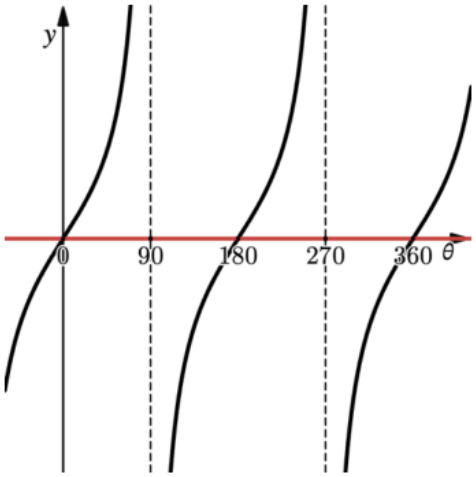
$\cos\theta = -1$ for $0° \leq \theta < 360°$.




$\Rightarrow \theta = \{180°\}$.

**4 a)**

$\tan\theta = 0$ for $0° \leq \theta < 360°$.



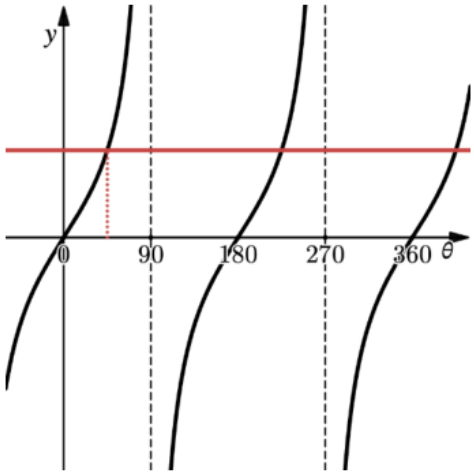

$\Rightarrow \theta = \{0°, 180°\}$.

**4 b)**

$\tan\theta = 1$ for $0° \leq \theta < 360°$.

Find first value:

$\theta = \arctan 1 = 45°$ (in range).




Add period:

$\theta = 45° + 180° = 225°$ (in range).

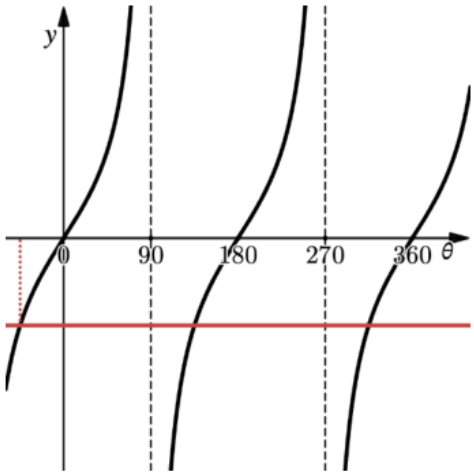
$\Rightarrow \theta = \{45°, 225°\}$.

**4 c)**

$\tan\theta = -1$ for $0° \leq \theta < 360°$.

Find first value:

$\theta = \arctan(-1) = -45°$ (not in range).



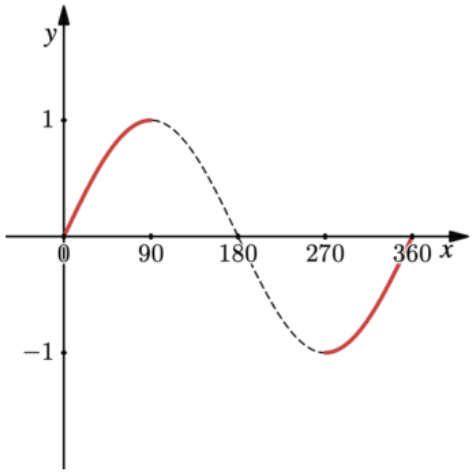

Add on period:

$\theta = -45° + 180° = 135°$ (in range).

$\theta = 135° + 180° = 315°$ (in range).

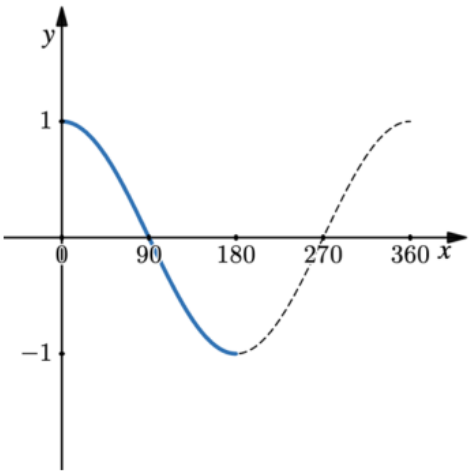
$\Rightarrow \theta = \{135°, 315°\}$.

**5)**




$0° < x < 90°$ or $270° < x < 360°$.

**6)**



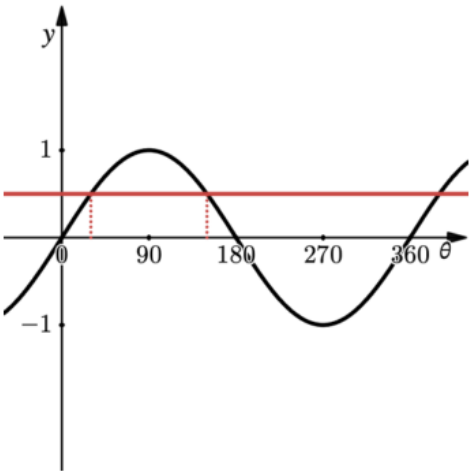

$0° < x < 180°$.

**7 a)**

$\sin\theta = \frac{1}{3}$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$\theta = \arcsin\frac{1}{3} = 19.471\ldots°$ (in range).




Find second value between $0°$ and $360°$:

$\theta = 180° - 19.471\ldots° = 160.528\ldots°$ (in range).

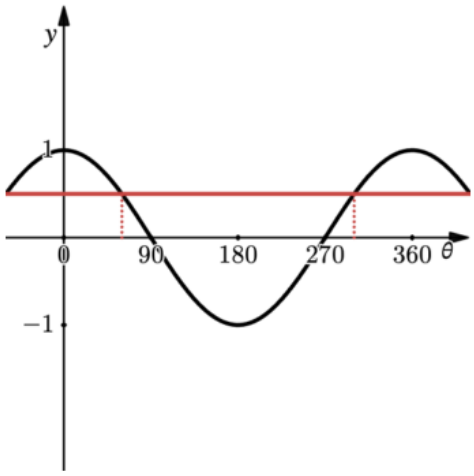
$\Rightarrow \theta = \{19.5°, 161°\}$ (3 SF).

**7 b)**

$\cos\theta = \frac{2}{3}$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$\theta = \arccos\frac{2}{3} = 48.189\ldots°$ (in range).




Find second value between $0°$ and $360°$:

$\theta = 360° - 48.189\ldots° = 311.810\ldots°$ (in range).

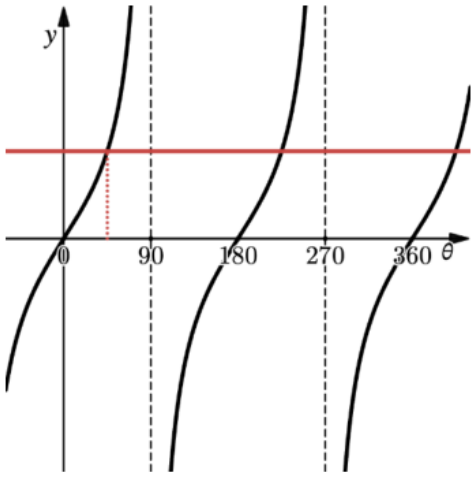
$\Rightarrow \theta = \{48.2°, 312°\}$ (3 SF).

**7 c)**

$\tan\theta = \frac{4}{3}$ for $0° \leq \theta < 360°$.

Find first value:

$\theta = \arctan\frac{4}{3} = 53.130\ldots°$ (in range).




Add period:

$\theta = 53.130\ldots° + 180° = 233.130\ldots°$ (in range).

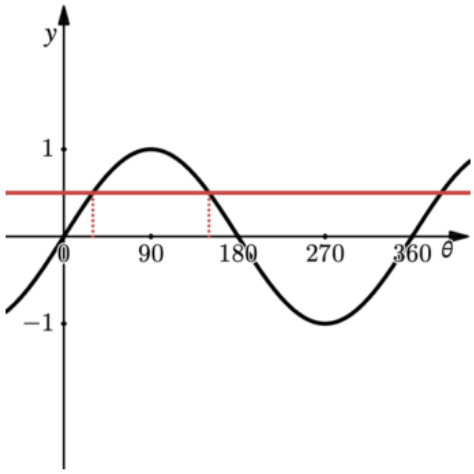
$\Rightarrow \theta = \{53.1°, 233°\}$ (3 SF).

**8 a)**

$\sin\theta = 0.4$ for $-360° \leq \theta < 0°$.

Find first value between $0°$ and $360°$:

$\theta = \arcsin 0.4 = 23.578\ldots°$ (not in range).




Find second value between $0°$ and $360°$:

$\theta = 180° - 23.578\ldots° = 156.421\ldots°$ (not in range).

Subtract period:

$\theta = 23.578\ldots° - 360° = -336.421\ldots°$ (in range).

$\theta = 156.421\ldots° - 360° = -203.578\ldots°$ (in range).

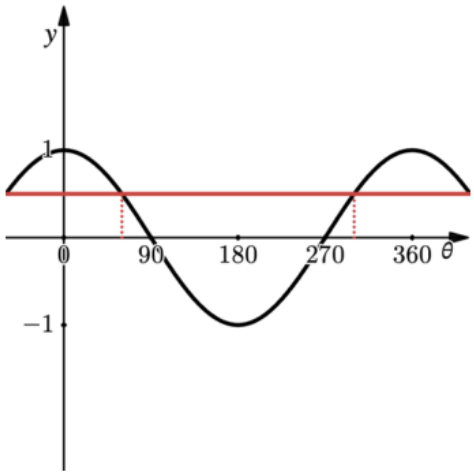
$\theta = \{-336°, -204°\}$ (3 SF).

**8 b)**

$\cos\theta = \frac{3}{7}$ for $-360° \leq \theta < 0°$.

Find first value between 0° and 360°:

$\theta = \arccos\frac{3}{7} = 64.623\ldots°$ (not in range).




Find second value between 0° and 360°:

$\theta = 360° - 64.623\ldots° = 295.376\ldots°$ (not in range).

Subtract period:

$\theta = 64.623\ldots° - 360° = -295.376\ldots°$ (in range).

$\theta = 295.376\ldots° - 360° = -64.623\ldots°$ (in range).

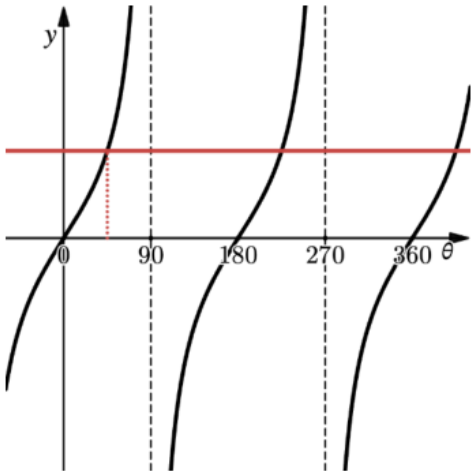
$\Rightarrow \theta = \{-295°, -64.6°\}$ (3 SF).

**8 c)**

$\tan\theta = \frac{4}{9}$ for $-360° \leq \theta < 0°$.

Find first value:

$\theta = \arctan\frac{4}{9} = 23.962\ldots°$ (not in range).




Subtract period:

$\theta = 23.962\ldots° - 180° = -156.037\ldots°$ (in range).

$\theta = -156.037\ldots° - 180° = -336.037\ldots°$ (in range).

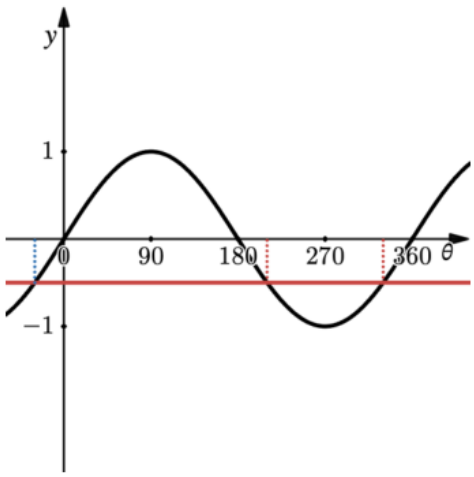
$\theta = \{-336°, -156°\}$ (3 SF).

**9 a)**

$\sin\theta = -0.2$ for $0° \leq \theta < 360°$.

Find first value:

$\theta = \arcsin(-0.2) = -11.536\ldots°$ (not in range).




Find first value between 0° and 360°:

$\theta = 180° + 11.536\ldots° = 191.536\ldots°$ (in range).

Find second value between 0° and 360°:

$\theta = 360° - 11.536\ldots° = 348.463\ldots°$ (in range).

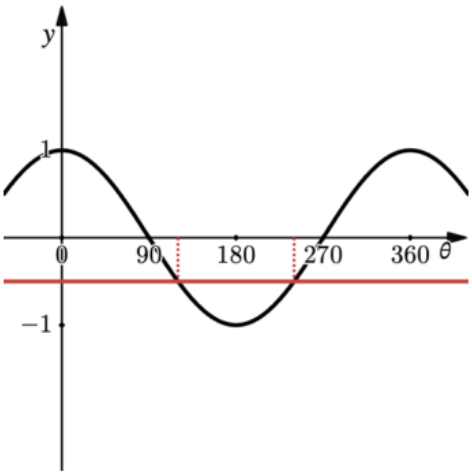
$\theta = \{192°, 348°\}$ (3 SF).

**9 b)**

$\cos\theta = -\frac{4}{7}$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$\theta = \arccos(-\frac{4}{7}) = 124.849\ldots°$ (in range).




Find second value between $0°$ and $360°$:

$\theta = 360° - 124.849\ldots° = 235.150\ldots°$ (in range).

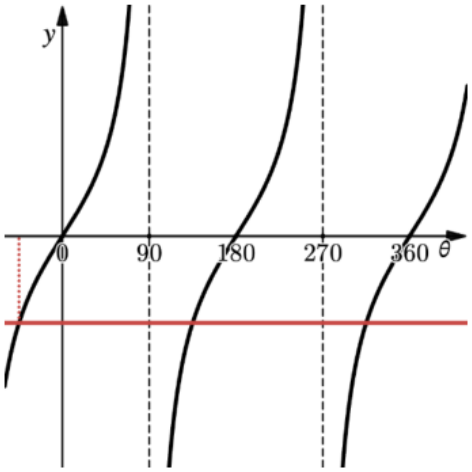
$\Rightarrow \theta = \{125°, 235°\}$ (3 SF).

**9 c)**

$\tan\theta = -4$ for $0° \leq \theta < 360°$.

Find first value:

$\theta = \arctan(-4) = -75.963\ldots°$ (not in range).




Add on period:

$\theta = -75.963\ldots° + 180° = 104.036\ldots°$ (in range).

$\theta = 104.036\ldots° + 180° = 284.036\ldots°$ (in range).

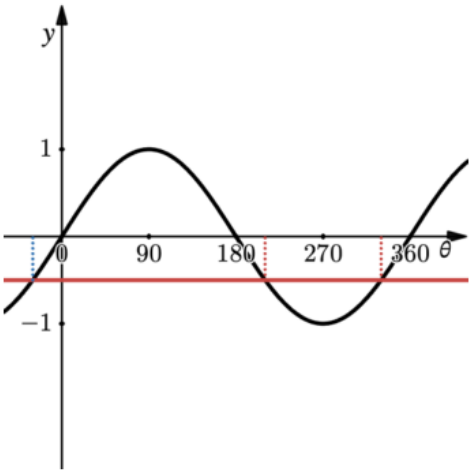
$\Rightarrow \theta = \{104°, 284°\}$ (3 SF).

**10 a)**

$\sin\theta = -0.9$ for $-360° \leq \theta < 0°$.

Find first value:

$\theta = \arcsin(-0.9) = -64.158\ldots°$ (in range).




Find first value between 0° and 360°:

$\theta = 180° + 64.158\ldots° = 244.158\ldots°$ (not in range).

Find second value between 0° and 360°:

$\theta = 360° - 64.158\ldots° = 295.841\ldots°$ (not in range).

Subtract period:

$\theta = 244.158\ldots° - 360° = -115.841\ldots°$ (in range).

$\theta = 295.841\ldots° - 360° = -64.158\ldots°$ (in range).

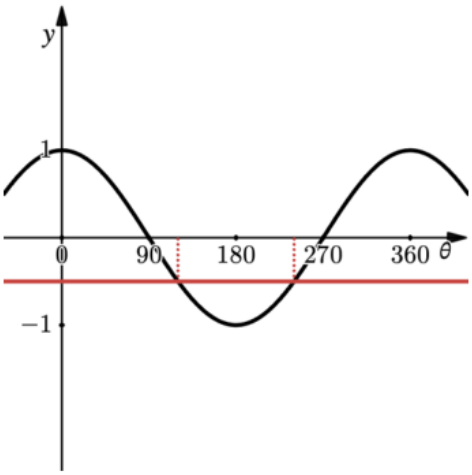
$\theta = \{-116°, -64.2°\}$ (3 SF).

**10 b)**

$\cos\theta = -\frac{3}{5}$ for $-360° \leq \theta < 0°$.

Find first value between 0° and 360°:

$\theta = \arccos(-\frac{3}{5}) = 126.869\ldots°$ (not in range).




Find second value between 0° and 360°:

$\theta = 360° - 126.869\ldots° = 233.130\ldots°$ (not in range).

Subtract period:

$\theta = 126.869\ldots° - 360° = -233.130\ldots°$ (in range).

$\theta = 233.130\ldots° - 360° = -126.869\ldots°$ (in range).

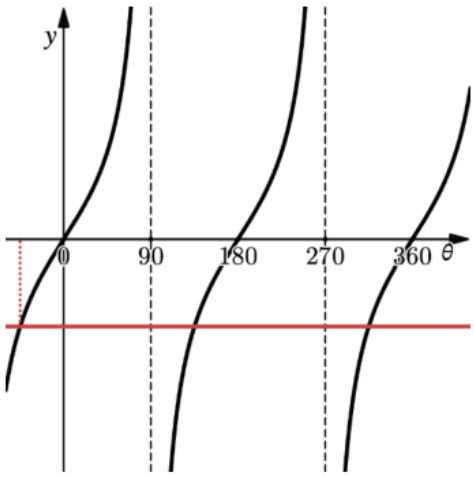
$\Rightarrow \theta = \{-233°, -127°\}$ (3 SF).

**10 c)**

$\tan\theta = -\frac{9}{2}$ for $-360° \leq \theta < 0°$.

Find first value:

$\theta = \arctan(-\frac{9}{2}) = -77.471\ldots°$ (in range).




Subtract period:

$\theta = -77.471\ldots° - 180° = -257.471\ldots°$ (in range).

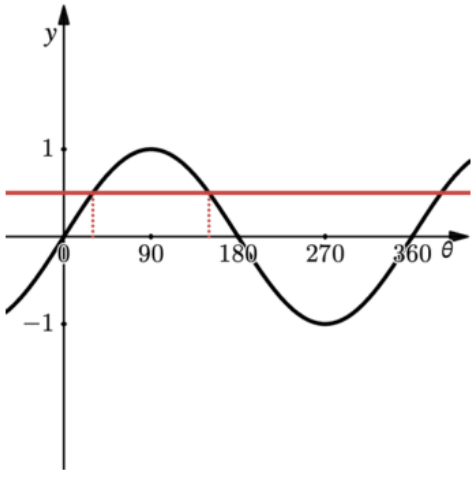
$\Rightarrow \theta = \{-257°, -77.5°\}$ (3 SF).

**11 a)**

$\sin\theta = \frac{3}{8}$ for $-360° \leq \theta < 360°$.

Find first value between 0° and 360°:

$\theta = \arcsin\frac{3}{8} = 22.024\ldots°$ (in range).




Find second value between 0° and 360°:

$\theta = 180° - 22.024\ldots° = 157.975\ldots°$ (in range).

Subtract period:

$\theta = 22.024\ldots° - 360° = -337.975\ldots°$ (in range).

$\theta = 157.975\ldots° - 360° = -202.024\ldots°$ (in range).

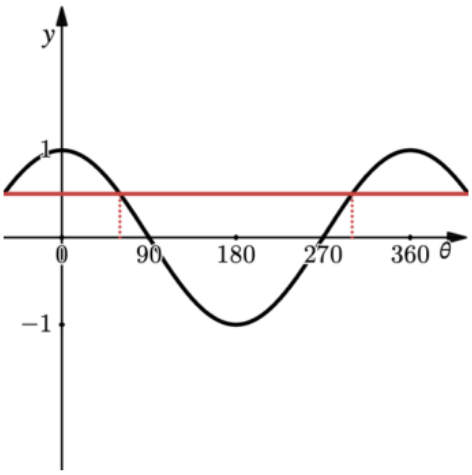
$\Rightarrow \theta = \{-338°, -202°, 22.0°, 158°\}$ (3 SF).

**11 b)**

$\cos\theta = 0.7$ for $-360° \leq \theta < 360°$.

Find first value between 0° and 360°:

$\theta = \arccos 0.7 = 45.572\ldots°$ (in range).




Find second value between 0° and 360°:

$\theta = 360° - 45.572\ldots° = 314.427\ldots°$ (in range).

Subtract period:

$\theta = 45.572\ldots° - 360° = -314.427\ldots°$ (in range).

$\theta = 314.427\ldots° - 360° = -45.572\ldots°$ (in range).

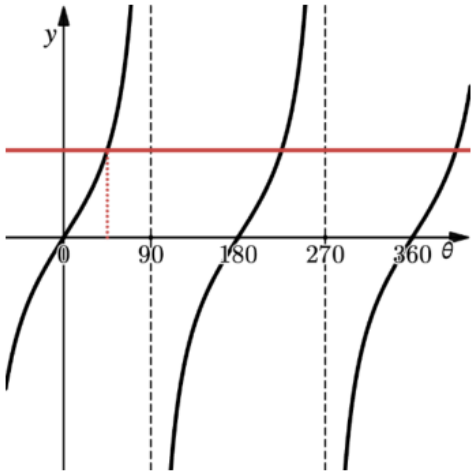
$\Rightarrow \theta = \{-314°, -45.6°, 45.6°, 314°\}$ (3 SF).

**11 c)**

$\tan\theta = 6$ for $-360° \leq \theta < 360°$.

Find first value:

$\theta = \arctan 6 = 80.537\ldots°$ (in range).




Add period:

$\theta = 80.537\ldots° + 180° = 260.537\ldots°$ (in range).

Subtract period:

$\theta = 80.537\ldots° - 180° = -99.462\ldots°$ (in range).

$\theta = -99.462\ldots° - 180° = -279.462\ldots°$ (in range).

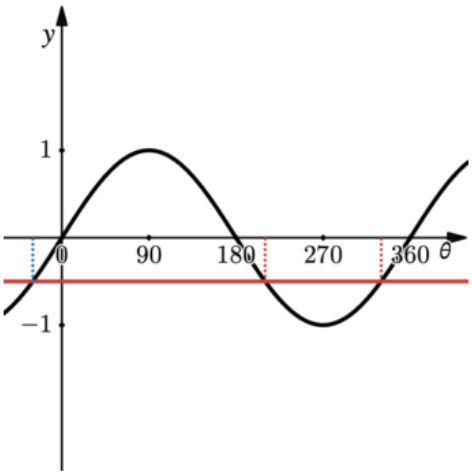
$\theta = \{-279°, -99.5°, 80.5°, 261°\}$ (3 SF).

**11 d)**

$\sin\theta = -\frac{6}{7}$ for $-360° \leq \theta < 360°$.

Find first value:

$\theta = \arcsin(-\frac{6}{7}) = -58.997\ldots°$ (in range).




Find first value between 0° and 360°:

$\theta = 180° + 58.997\ldots° = 238.997\ldots°$ (in range).

Find second value between 0° and 360°:

$\theta = 360° - 58.997\ldots° = 301.002\ldots°$ (in range).

Subtract period:

$\theta = 238.997\ldots° - 360° = -121.002\ldots°$ (in range).

$\theta = 301.002\ldots° - 360° = -58.997\ldots°$ (in range).

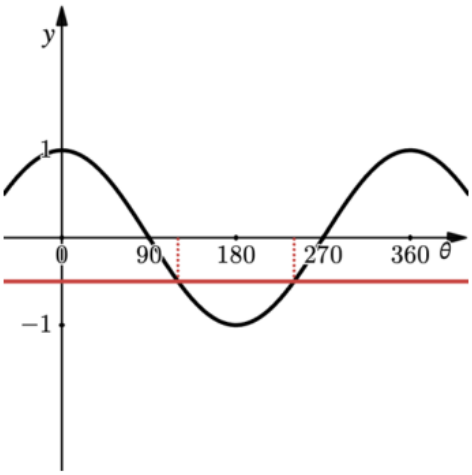
$\theta = \{-121°, -59.0°, 239°, 301°\}$ (3 SF).

**11 e)**

$\cos\theta = -\frac{2}{9}$ for $-360° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$\theta = \arccos(-\frac{2}{9}) = 102.839\ldots°$ (in range).




Find second value between $0°$ and $360°$:

$\theta = 360° - 102.839\ldots° = 257.160\ldots°$ (in range).

Subtract period:

$\theta = 102.839\ldots° - 360° = -257.160\ldots°$ (in range).

$\theta = 257.160\ldots° - 360° = -102.839\ldots°$ (in range).

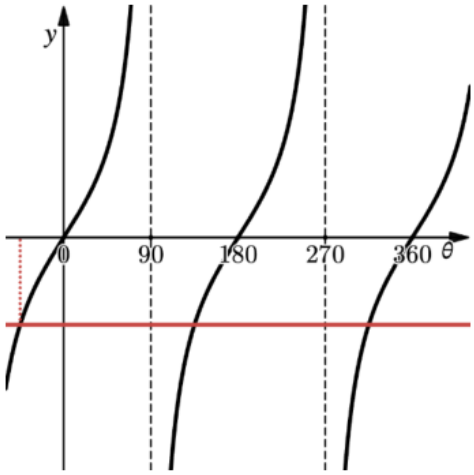
$\Rightarrow \theta = \{-257°, -103°, 103°, 257°\}$ (3 SF).

**11 f)**

$\tan\theta = -1.3$ for $-360° \leq \theta < 360°$.

Find first value:

$\theta = \arctan(-1.3) = -52.431\ldots°$ (in range).




Add period:

$\theta = -52.431\ldots° + 180° = 127.568\ldots°$ (in range).

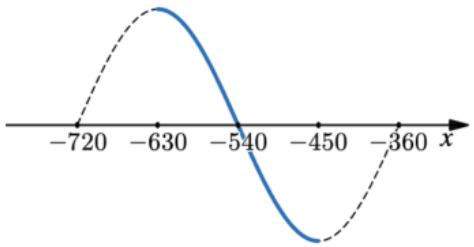
$\theta = 127.568\ldots° + 180° = 307.568\ldots°$ (in range).

Subtract period:

$\theta = -52.431\ldots° - 180° = -232.431\ldots°$ (in range).

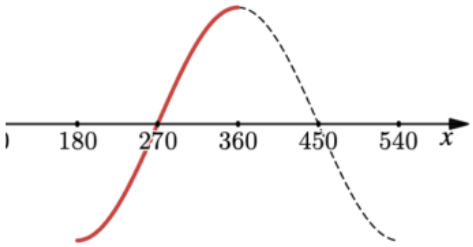
$\Rightarrow \theta = \{-232°, -52.4°, 128°, 308°\}$ (3 SF).

**12)**




$-630° < x < -450°$.

**13)**



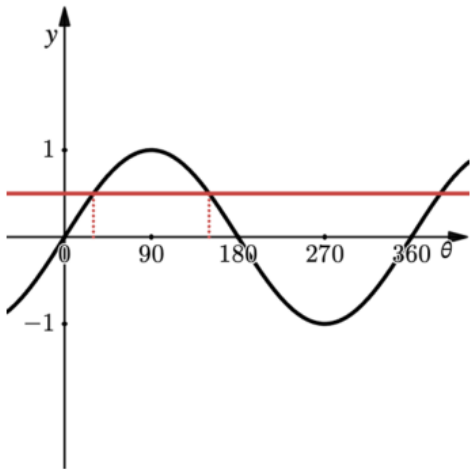

$180° < x < 360°$.

**14 a)**

$\sin\theta = \frac{4}{7}$ for $0° \leq \theta < 720°$.

Find first value between $0°$ and $360°$:

$\theta = \arcsin\frac{4}{7} = 34.849\ldots°$ (in range).




Find second value between $0°$ and $360°$:

$\theta = 180° - 34.849\ldots° = 145.150\ldots°$ (in range).

Add period:

$\theta = 34.849\ldots° + 360° = 394.849\ldots°$ (in range).

$\theta = 145.150\ldots° + 360° = 505.150\ldots°$ (in range).

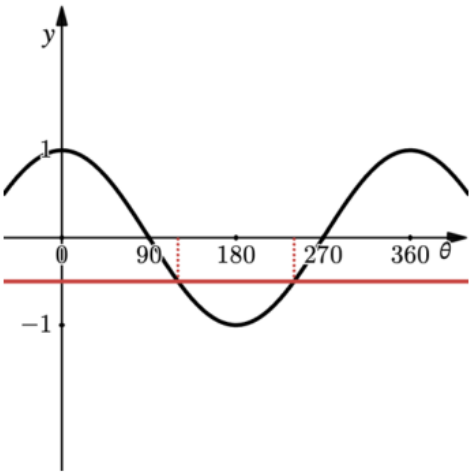
$\theta = \{34.8°, 145°, 395°, 505°\}$ (3 SF).

**14 b)**

$\cos\theta = -\frac{8}{9}$ for $720° \leq \theta < 1080°$.

Find first value between $0°$ and $360°$:

$\theta = \arccos(-\frac{8}{9}) = 152.733\ldots°$ (not in range).




Find second value between $0°$ and $360°$:

$\theta = 360° - 152.733\ldots° = 207.266\ldots°$ (not in range).

Add period:

$\theta = 152.733\ldots° + 360° = 512.733\ldots°$ (not in range).

$\theta = 512.733\ldots° + 360° = 872.733\ldots°$ (in range).

$\theta = 207.266\ldots° + 360° = 567.266\ldots°$ (not in range).

$\theta = 567.266\ldots° + 360° = 927.266\ldots°$ (in range).

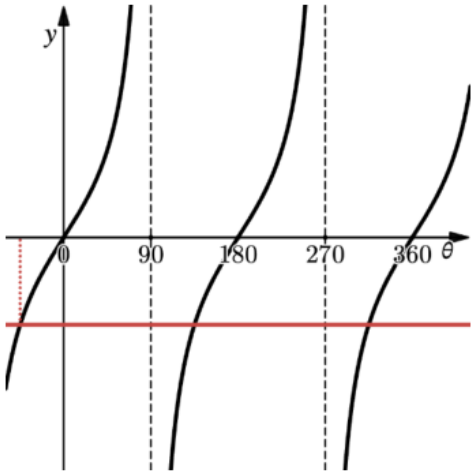
$\Rightarrow \theta = \{873°, 927°\}$ (3 SF).

**14 c)**

$\tan\theta = -9$ for $-720° \leq \theta < -540°$.

Find first value:

$\theta = \arctan(-9) = -83.659\ldots°$ (not in range).




Subtract period:

$\theta = -83.659\ldots° - 180° = -263.659\ldots°$ (not in range).

$\theta = -263.659\ldots° - 180° = -443.659\ldots°$ (not in range).

$\theta = -443.659\ldots° - 180° = -623.659\ldots°$ (in range).

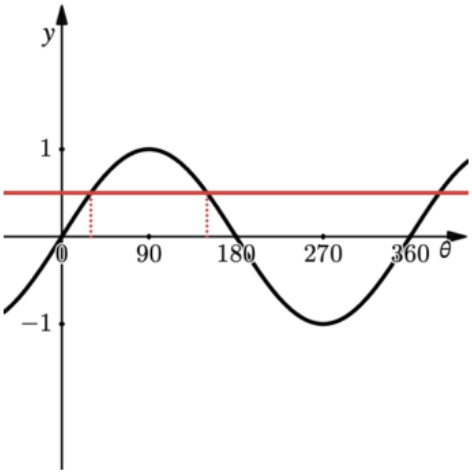
$\Rightarrow \theta = \{-624°\}$ (3 SF).

**15 a)**

$\frac{1}{\sin\theta} = 18 \Rightarrow \sin\theta = \frac{1}{18}$ for $0° \leq \theta < 180°$.

Find first value between 0° and 360°:

$\theta = \arcsin\frac{1}{18} = 3.184\ldots°$ (in range).




Find second value between 0° and 360°:

$\theta = 180° - 3.184\ldots° = 176.815\ldots°$ (in range).

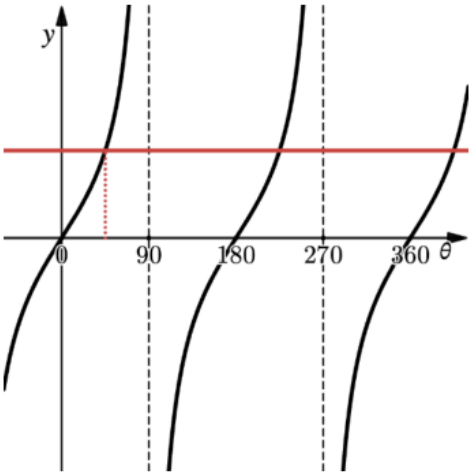
$\theta = \{3.18°, 177°\}$ (3 SF).

**15 b)**

$\frac{5}{\tan\theta} = 13 \Rightarrow \tan\theta = \frac{5}{13}$ for $180° \leq \theta < 540°$.

Find first value:

$\theta = \arctan\frac{5}{13} = 21.037\ldots°$ (not in range).




Add period:

$\theta = 21.037\ldots° + 180° = 201.037\ldots°$ (in range).

$\theta = 201.037\ldots° + 180° = 381.037\ldots°$ (in range).

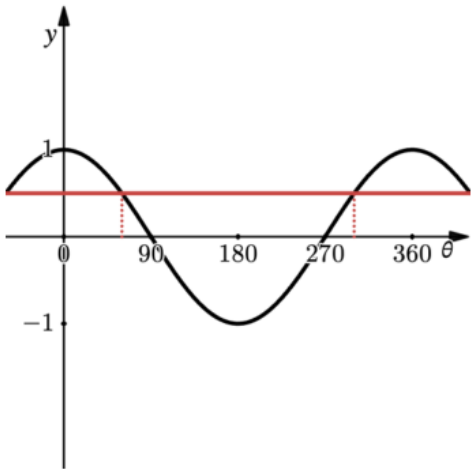
$\theta = \{201°, 381°\}$ (3 SF).

**15 c)**

$\frac{0.1}{\cos\theta} = 2 \Rightarrow \cos\theta = 0.05$ for $-180° \leq \theta < 180°$.

Find first value between 0° and 360°:

$\theta = \arccos 0.05 = 87.134\ldots°$ (in range).




Find second value between 0° and 360°:

$\theta = 360° - 87.134\ldots° = 272.865\ldots°$ (not in range).

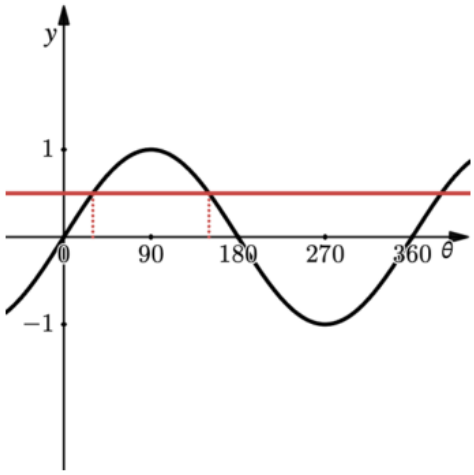
Subtract period:

$\theta = 272.865\ldots° - 360° = -87.134\ldots°$ (in range).

$\Rightarrow \theta = \{-87.1°, 87.1°\}$ (3 SF).

**16)**

$\frac{\sin A}{18} = \frac{\sin 20°}{15} \Rightarrow A = \arcsin(\frac{18 \sin 20°}{15}) = 24.231\ldots$ (in range).




Find second value between 0° and 360°:

$A = 180° - 24.231\ldots° = 155.768\ldots°$ (in range).

$A = \{24.2°, 156°\}$ (3 SF).

**17)**

$A : x = p.$

$B : x = 180° - p.$

$C : x = -180° - p.$

$D : x = p - 360°.$

**18)**

$\sin 60° = \frac{\sqrt{3}}{2}.$

$\sin(180° - 60°) = \sin 120° = \frac{\sqrt{3}}{2}.$

$\sin(120° + 360°) = \sin 480° = \frac{\sqrt{3}}{2}.$

**19)**

$\cos 45° = \frac{\sqrt{2}}{2}.$

$\cos(90° + 45°) = \cos 135° = -\frac{\sqrt{2}}{2}.$

$\cos(135° + 360°) = \cos 495° = -\frac{\sqrt{2}}{2}.$

**20)**

$\tan(-210°) = -\frac{\sqrt{3}}{3}.$

$\tan 210° = \frac{\sqrt{3}}{3}.$

$\tan(210° + 180°) = \tan 390° = \frac{\sqrt{3}}{3}.$

**21)**

$A : x = p.$

$B : x = 180° - p.$

$C : x = p - 360°.$

$D : x = -180° - p.$

**22)**

$A : x = -p.$

$B : x = -180° + p.$

$C : x = 180° + p.$

$D : x = 360° - p.$

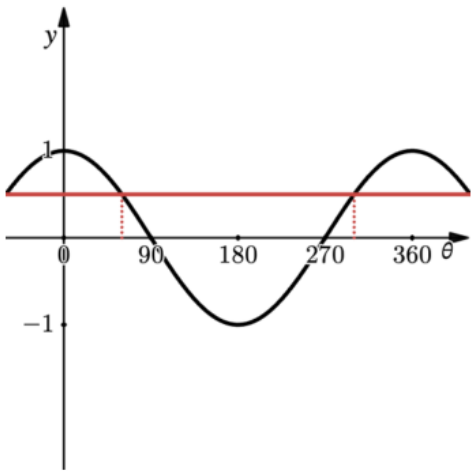
**23 a)**

$\cos^2\theta = \frac{1}{16} \Rightarrow \cos\theta = \pm\frac{1}{4}$ for $0° \leq \theta < 360°$.

Solve $\cos\theta = \frac{1}{4}$ first:

Find first value between 0° and 360°:

$\theta = \arccos\frac{1}{4} = 75.522\ldots°$ (in range).



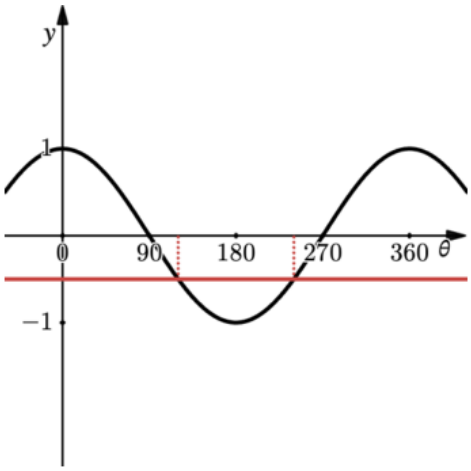

Find second value between 0° and 360°:

$\theta = 360° - 75.522\ldots° = 284.477\ldots°$ (in range).

Now solve $\cos\theta = -\frac{1}{4}$:

Find first value between 0° and 360°:

$\theta = \arccos(-\frac{1}{4}) = 104.477\ldots°$ (in range).




Find second value between 0° and 360°:

$\theta = 360° - 104.477\ldots° = 255.522\ldots°$ (in range).

$\Rightarrow \theta = \{75.5°, 104°, 256°, 284°\}$ (3 SF).

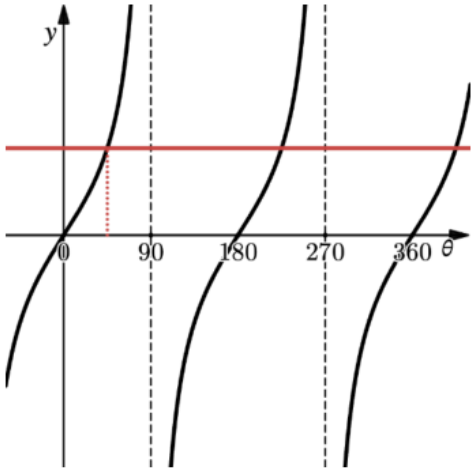
**23 b)**

$3\tan^2\theta = 0.8 \Rightarrow \tan\theta = \pm\frac{2\sqrt{15}}{15}$ for $-180° \leq \theta < 180°$.

Solve $\tan\theta = \frac{2\sqrt{15}}{15}$ first:

Find first value:

$\theta = \arctan\frac{2\sqrt{15}}{15} = 27.311\ldots°$ (in range).



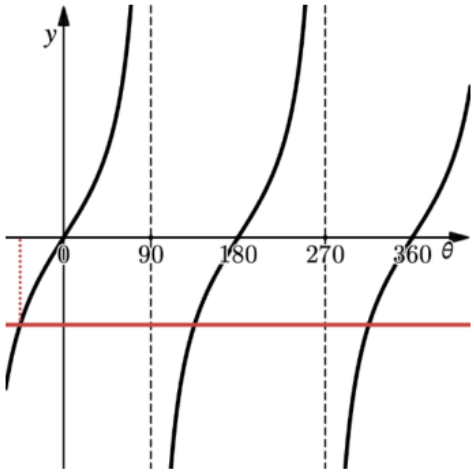

Subtract period:

$\theta = 27.311\ldots° - 180° = -152.688\ldots°$ (in range).

Now solve $\tan\theta = -\frac{2\sqrt{15}}{15}$:

Find first value:

$\theta = \arctan(-\frac{2\sqrt{15}}{15}) = -27.311\ldots°$ (in range).




Add period:

$\theta = -27.311\ldots^\circ + 180^\circ = 152.688\ldots^\circ$ (in range).

$\theta = \{-153^\circ, -27.3^\circ, 27.3^\circ, 153^\circ\}$ (3 SF).

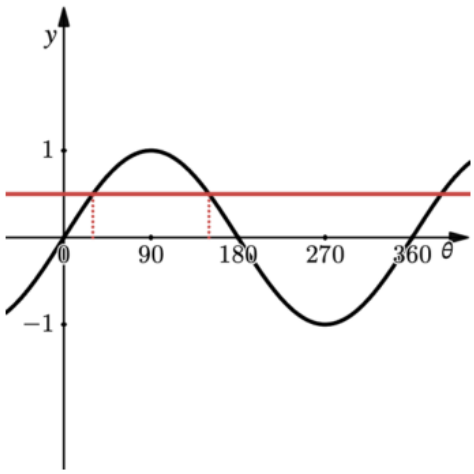
**23 c)**

$10\sin^2\theta = 7 \Rightarrow \sin\theta = \pm\sqrt{\frac{7}{10}}$ for $0^\circ \leq \theta < 540^\circ$.

Solve $\sin\theta = \sqrt{\frac{7}{10}}$ first:

Find first value between $0^\circ$ and $360^\circ$:

$\theta = \arcsin\sqrt{\frac{7}{10}} = 56.789\ldots^\circ$ (in range).




Find second value between $0^\circ$ and $360^\circ$:

$\theta = 180^\circ - 56.789\ldots^\circ = 123.210\ldots^\circ$ (in range).

Add period:

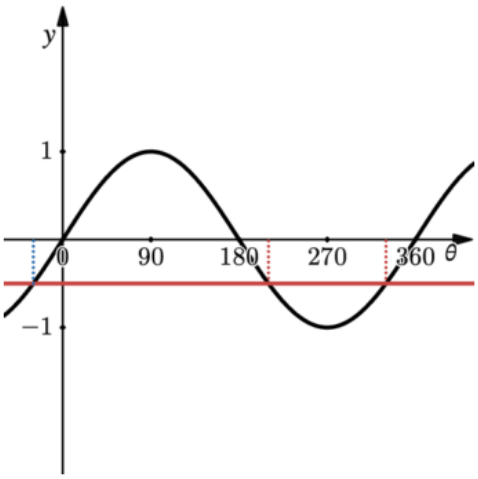
$\theta = 56.789\ldots^\circ + 360^\circ = 416.789\ldots^\circ$ (in range).

$\theta = 123.210\ldots^\circ + 360^\circ = 483.210\ldots^\circ$ (in range).

Now solve $\sin\theta = -\sqrt{\frac{7}{10}}$:

Find first value:

$\theta = \arcsin(-\sqrt{\frac{7}{10}}) = -56.789\ldots^\circ$ (not in range).




Find first value between 0° and 360°:

$\theta = 180° + 56.789\ldots° = 236.789\ldots°$ (in range).

Find second value between 0° and 360°:

$\theta = 360° - 56.789\ldots° = 303.210\ldots°$ (in range).

$\theta = \{56.8°, 123°, 237°, 303°, 417°, 483°\}$ (3 SF).

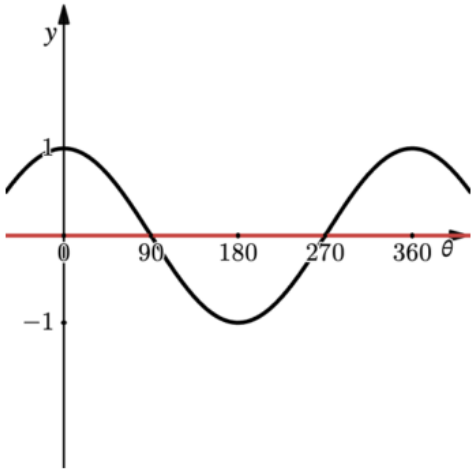
**24 a)**

$\cos^2\theta = 2\cos\theta$ for $0° \leq \theta < 360°$.

$\Rightarrow \cos^2\theta - 2\cos\theta = 0 \Rightarrow \cos\theta(\cos\theta - 2) = 0$.

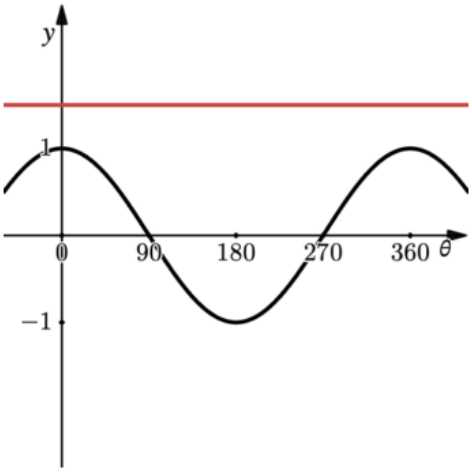
$\Rightarrow \cos\theta = 0$ or $\cos\theta = 2$.

Solve $\cos\theta = 0$ first:




$\theta = 90°$ or $\theta = 270°$.

As for $\cos\theta = 2$, there are no solutions:




$\theta = \{90°, 270°\}$.

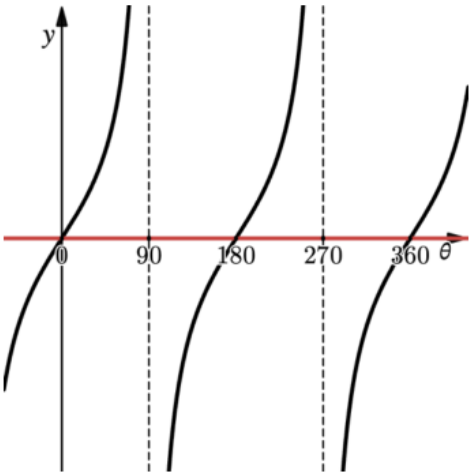
**24 b)**

$7\tan^2\theta + 4\tan\theta = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow \tan\theta(7\tan\theta + 4) = 0$.

$\Rightarrow \tan\theta = 0$ or $\tan\theta = -\frac{4}{7}$.

Solve $\tan\theta = 0$ first:

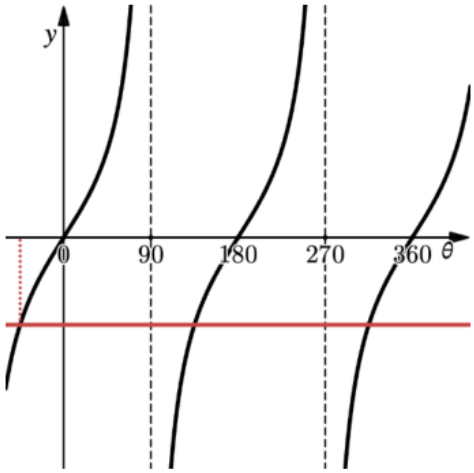



$\theta = 0°$ or $\theta = 180°$.

Now solve $\tan\theta = -\frac{4}{7}$:

Find first value:

$\theta = \arctan(-\frac{4}{7}) = -29.744\ldots°$ (not in range).




Add period:

$\theta = -29.744\ldots^\circ + 180^\circ = 150.255\ldots^\circ$ (in range).

$\theta = 150.255\ldots^\circ + 180^\circ = 330.255\ldots^\circ$ (in range).

$\theta = \{0^\circ, 150^\circ, 180^\circ, 330^\circ\}$ (3 SF).

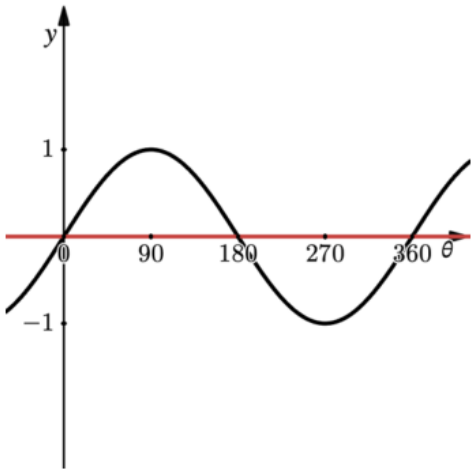
**24 c)**

$2\sin\theta = 5\sin^2\theta$ for $0^\circ \leq \theta < 360^\circ$.

$2\sin\theta - 5\sin^2\theta = 0 \Rightarrow \sin\theta(2 - 5\sin\theta) = 0$.

$\Rightarrow \sin\theta = 0$ or $\sin\theta = \frac{2}{5}$.

Solve $\sin\theta = 0$ first:

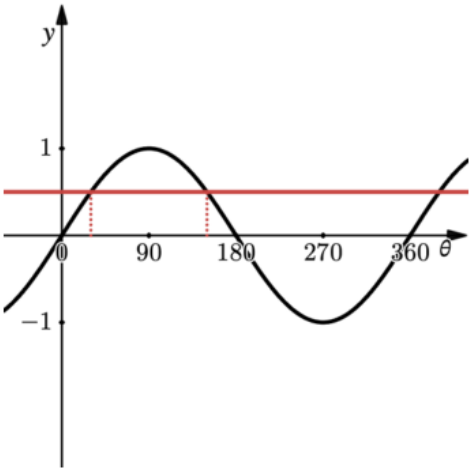



$\theta = 0^\circ$ or $\theta = 180^\circ$.

Now solve $\sin\theta = \frac{2}{5}$:

Find first value between $0^\circ$ and $360^\circ$:

$\theta = \arcsin\frac{2}{5} = 23.578\ldots^\circ$ (in range).




Find second value between $0°$ and $360°$:

$\theta = 180° - 23.578\ldots° = 156.421\ldots°$ (in range).

$\theta = \{0°, 23.6°, 156°, 180°\}$ (3 SF).

**25)**

$A : x = p.$

$B : x = 360° - p.$

$C : x = -p.$

$D : x = -360° + p.$

**26)**

$A : x = p.$

$B : x = 360° - p.$

$C : x = p - 360°.$

$D : x = -p.$

**27)**

$A : x = p.$

$B : x = p + 180°.$

$C : x = p - 180°.$

$D : x = p - 360°.$

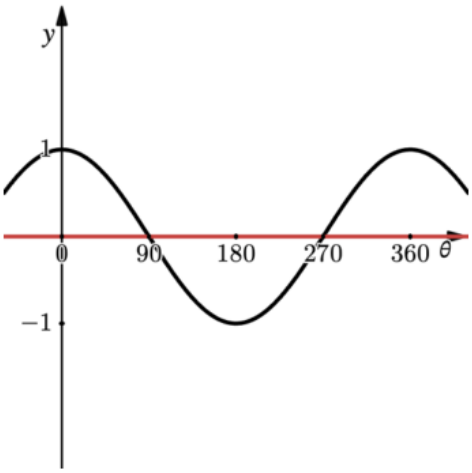
**28 a)**

$3\sin\theta\cos\theta = \cos\theta$ for $0° \leq \theta < 360°$.

$\Rightarrow 3\sin\theta\cos\theta - \cos\theta = 0.$

$\Rightarrow \cos\theta(3\sin\theta - 1) = 0 \Rightarrow \cos\theta = 0$ or $\sin\theta = \frac{1}{3}$.

Solve $\cos\theta = 0$ first:

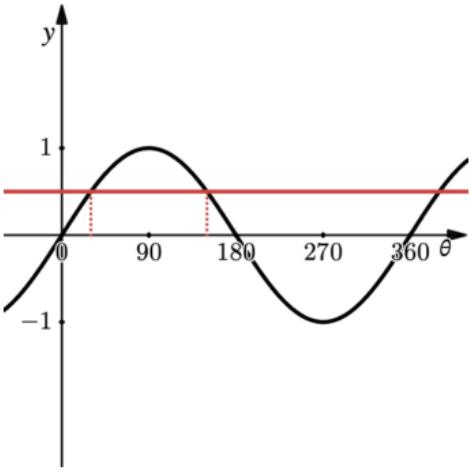



$\theta = 90°$ or $\theta = 270°$.

Now solve $\sin\theta = \frac{1}{3}$:

Find first value between 0° and 360°:

$\theta = \arcsin\frac{1}{3} = 19.471\ldots°$ (in range).




Find second value between 0° and 360°:

$\theta = 180° - 19.471\ldots° = 160.528\ldots°$ (in range).

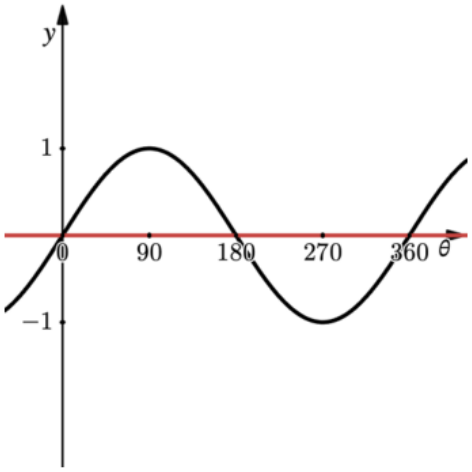
$\theta = \{19.5°, 90°, 161°, 270°\}$ (3 SF).

**28 b)**

$8\sin\theta\tan\theta - 9\sin\theta = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow \sin\theta(8\tan\theta - 9) = 0 \Rightarrow \sin\theta = 0$ or $\tan\theta = \frac{9}{8}$.

Solve $\sin\theta = 0$ first:

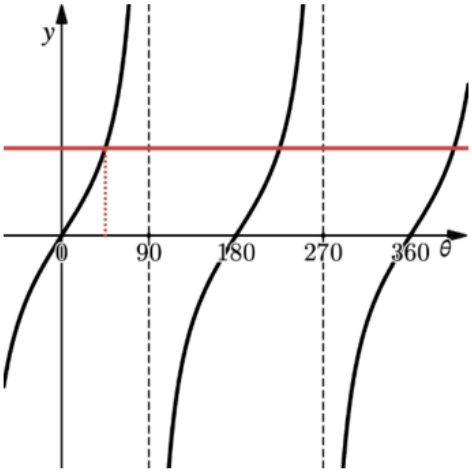



$\theta = 0°$ or $\theta = 180°$.

Now solve $\tan\theta = \frac{9}{8}$:

Find first value:

$\theta = \arctan\frac{9}{8} = 48.366\ldots°$ (in range).




Add period:

$\theta = 48.366\ldots° + 180° = 228.366\ldots°$ (in range).

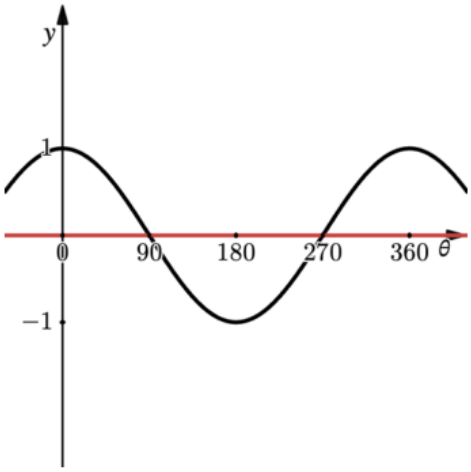
$\theta = \{0°, 48.4°, 180°, 228°\}$ (3 SF).

**28 c)**

$6\cos\theta - 5\sin\theta\cos\theta = 0$ for $0° \leq \theta < 360°$.

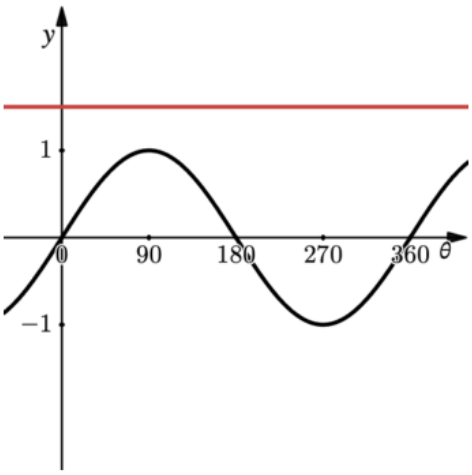
$\Rightarrow \cos\theta(6 - 5\sin\theta) = 0 \Rightarrow \cos\theta = 0$ or $\sin\theta = \frac{6}{5}$.

Solve $\cos\theta = 0$ first:




$\theta = 90°$ or $\theta = 270°$.

As for $\sin\theta = \frac{6}{5}$, there are no solutions:




$\theta = \{90°, 270°\}$.

**29 a)**

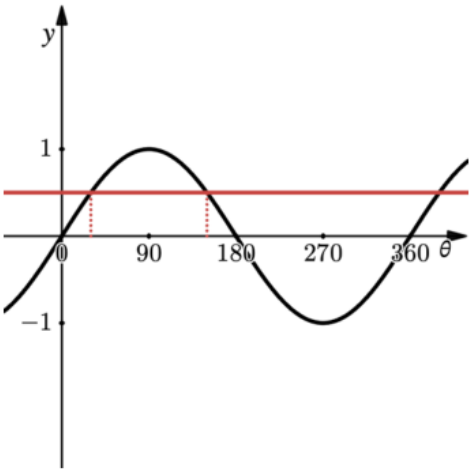
$2\sin^2\theta + \sin\theta - 1 = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow (2\sin\theta - 1)(\sin\theta + 1) = 0 \Rightarrow \sin\theta = \frac{1}{2}$ or $\sin\theta = -1$.

Solve $\sin\theta = \frac{1}{2}$ first:

Find first value between 0° and 360°:

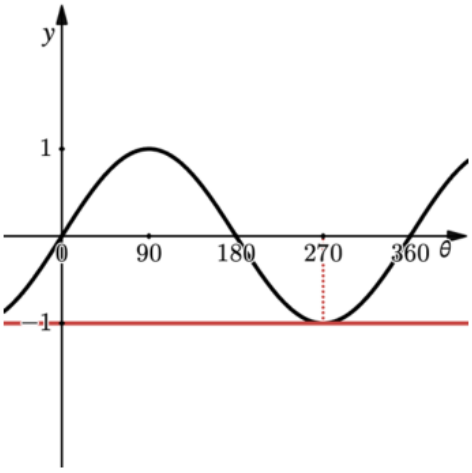
$\theta = \arcsin\frac{1}{2} = 30°$ (in range).




Find second value between 0° and 360°:

$\theta = 180° - 30° = 150°$ (in range).

Now solve $\sin\theta = -1$:




$\theta = 270°$.

$\theta = \{30°, 150°, 270°\}$.

**29 b)**

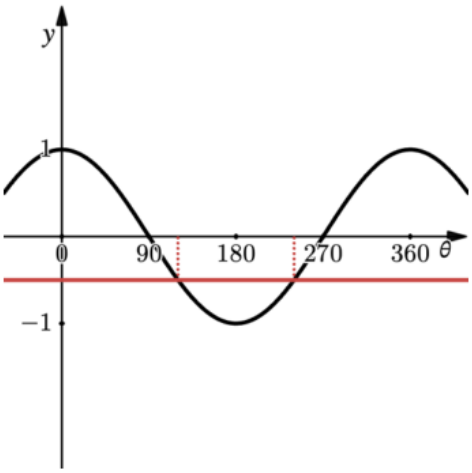
$3\cos^2\theta - \cos\theta - 2 = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow (3\cos\theta + 2)(\cos\theta - 1) = 0 \Rightarrow \cos\theta = -\frac{2}{3}$ or $\cos\theta = 1$.

Solve $\cos\theta = -\frac{2}{3}$ first:

Find first value between $0°$ and $360°$:

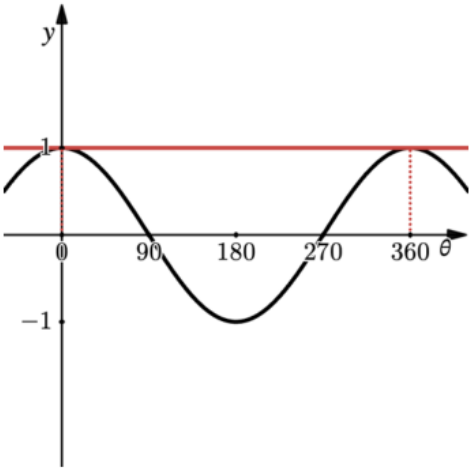
$\theta = \arccos(-\frac{2}{3}) = 131.810\ldots°$ (in range).




Find second value between $0°$ and $360°$:

$\theta = 360° - 131.810\ldots° = 228.189\ldots°$ (in range).

Now solve $\cos\theta = 1$:




$\theta = 0°$.

$\theta = \{0°, 132°, 228°\}$ (3 SF).

**29 c)**

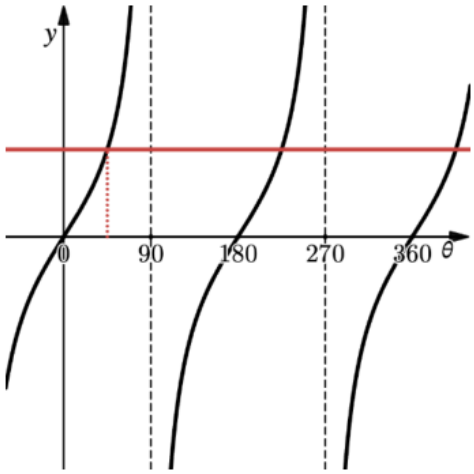
$4\tan^2\theta + 5\tan\theta - 6 = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow (4\tan\theta - 3)(\tan\theta + 2) = 0 \Rightarrow \tan\theta = \frac{3}{4}$ or $\tan\theta = -2$.

Solve $\tan\theta = \frac{3}{4}$ first:

Find first value:

$\theta = \arctan\frac{3}{4} = 36.869\ldots°$ (in range).



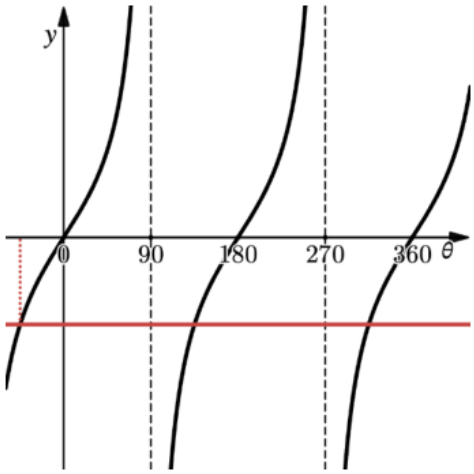

Add period:

$\theta = 36.869\ldots° + 180° = 216.869\ldots°$ (in range).

Now solve $\tan\theta = -2$:

Find first value:

$\theta = \arctan(-2) = -63.434\ldots°$ (not in range).




Add period:

$\theta = -63.434\ldots° + 180° = 116.565\ldots°$ (in range).

$\theta = 116.565\ldots° + 180° = 296.565\ldots°$ (in range).

$\theta = \{36.9°, 117°, 217°, 297°\}$ (3 SF).

**30 a)**

$6\sin\theta = 1 + \frac{1}{\sin\theta}$ for $0° \leq \theta < 360°$.

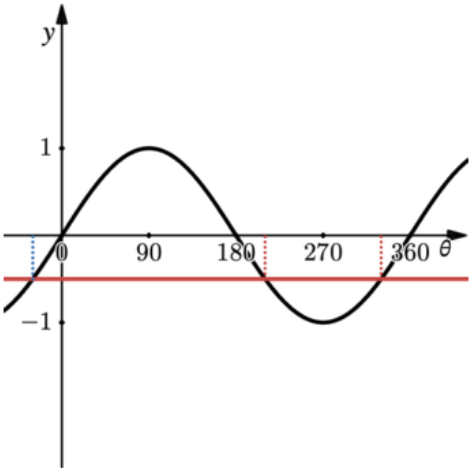
$\Rightarrow 6\sin^2\theta = \sin\theta + 1 \Rightarrow 6\sin^2\theta - \sin\theta - 1 = 0.$

$\Rightarrow (3\sin\theta + 1)(2\sin\theta - 1) = 0 \Rightarrow \sin\theta = -\frac{1}{3}$ or $\sin\theta = \frac{1}{2}$.

Solve $\sin\theta = -\frac{1}{3}$ first:

Find first value:

$\theta = \arcsin(-\frac{1}{3}) = -19.471\ldots°$ (not in range).




Find first value between 0° and 360°:

$\theta = 180° + 19.471\ldots° = 199.471\ldots°$ (in range).

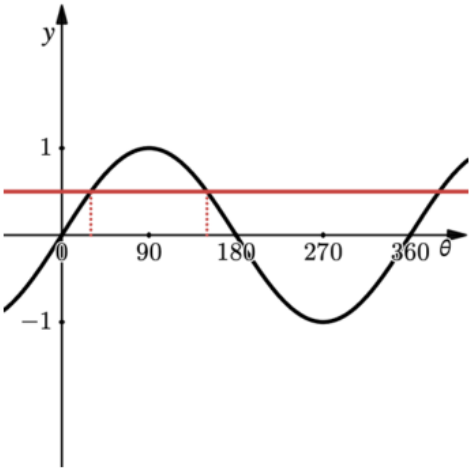
Find second value between 0° and 360°:

$\theta = 360° - 19.471\ldots° = 340.528\ldots°$ (in range).

Now solve $\sin\theta = \frac{1}{2}$:

Find first value between 0° and 360°:

$\theta = \arcsin\frac{1}{2} = 30°$ (in range).




Find second value between 0° and 360°: $\theta = 180° - 30° = 150°$ (in range).

$\theta = \{30°, 150°, 199°, 341°\}$ (3 SF).

**30 b)**

$9\cos\theta = \frac{16}{\cos\theta} - 32 \Rightarrow 9\cos^2\theta = 16 - 32\cos\theta.$

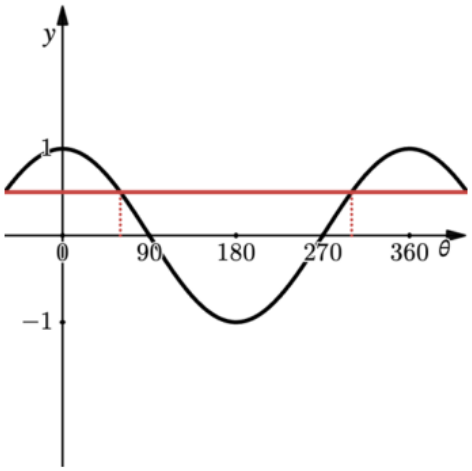
$\Rightarrow 9\cos^2\theta + 32\cos\theta - 16 = 0.$

$\Rightarrow (9\cos\theta - 4)(\cos\theta + 4) = 0 \Rightarrow \cos\theta = \frac{4}{9}$ or $\cos\theta = -4.$

Solve $\cos\theta = \frac{4}{9}$ first:

Find first value between 0° and 360°:

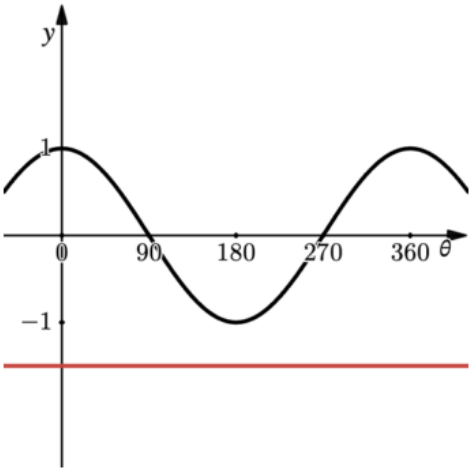
$\theta = \arccos\frac{4}{9} = 63.612\ldots°$ (in range).




Find second value between 0° and 360°:

$\theta = 360° - 63.612\ldots° = 296.387\ldots°$ (in range).

As for $\cos\theta = -4$, there are no solutions:




$\theta = \{63.6°, 296°\}$ (3 SF).

**30 c)**

$30\tan\theta = 23 + \frac{40}{\tan\theta} \Rightarrow 30\tan^2\theta = 23\tan\theta + 40.$

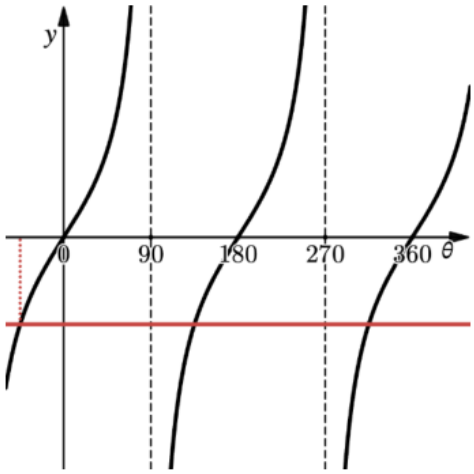
$\Rightarrow 30\tan^2\theta - 23\tan\theta - 40 = 0.$

$\Rightarrow (6\tan\theta + 5)(5\tan\theta - 8) = 0 \Rightarrow \tan\theta = -\frac{5}{6}$ or $\tan\theta = \frac{8}{5}$.

Solve $\tan\theta = -\frac{5}{6}$ first:

Find first value:

$\theta = \arctan(-\frac{5}{6}) = -39.805\ldots^\circ$ (not in range).




Add period:

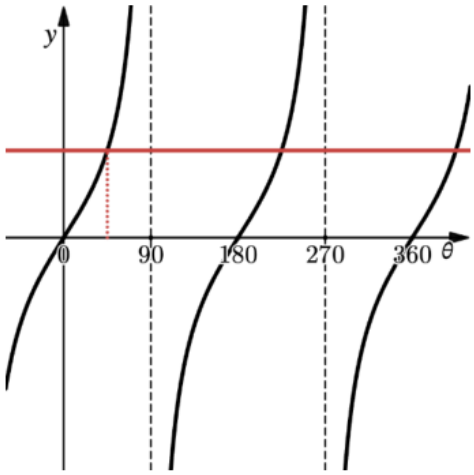
$\theta = -39.805\ldots^\circ + 180^\circ = 140.194\ldots^\circ$ (in range).

$\theta = 140.194\ldots^\circ + 180^\circ = 320.194\ldots^\circ$ (in range).

Now solve $\tan\theta = \frac{8}{5}$:

Find first value:

$\theta = \arctan\frac{8}{5} = 57.994\ldots^\circ$ (in range).




Add period:

$\theta = 57.994\ldots^\circ + 180^\circ = 237.994\ldots^\circ$ (in range). $\theta = \{58.0^\circ, 140^\circ, 238^\circ, 320^\circ\}$ (3 SF).

**31 a)**

$$\begin{aligned} 3x^2 - 12x + 17 &\equiv 3(x^2 - 4x) + 17 \\ &\equiv 3((x-2)^2 - 4) + 17 \equiv 3(x-2)^2 - 12 + 17 \\ &\equiv 3(x-2)^2 + 5 \end{aligned}$$

**31 b)**

$(2, 5)$.

**31 c)**

$3\sin^2\theta - 12\sin\theta + 17 \equiv 3(\sin\theta - 2)^2 + 5$.

The minimum value will occur when $(\sin\theta - 2)^2$ is at its minimum. $-1 \leq \sin\theta \leq 1$, so the minimum will occur when $\sin\theta = 1$, which gives $3(1-2)^2 + 5 = 8$. The smallest positive value of $\theta$ for which this minimum occurs is when $\theta = 90°$.

**32 a)**

Period is $360°$.

**32 b)**

Period is $360°$.

**32 c)**

Period is $180°$.

**32 d)**

Period is $\frac{360}{2} = 180°$.

**32 e)**

Period is $\frac{360}{3} = 120°$.

**32 f)**

Period is $\frac{180}{4} = 45°$.

**32 g)**

Period is $\frac{360}{\frac{1}{3}} = 1080°$.

**32 h)**

Period is $\frac{360}{\frac{2}{3}} = 540°$.

**32 i)**

Period is $\frac{180}{\frac{4}{5}} = 225°$.

**33 a)**

Period is $\frac{360}{2} = 180°$ and expect two solutions each period:
$\frac{360-0}{180} \times 2 = 4$ solutions.

**33 b)**

$\frac{180-0}{180} \times 2 = 2$ solutions.

**33 c)**

$\frac{720-0}{180} \times 2 = 8$ solutions.

**34 a)**

Period is $\frac{360}{2} = 180°$ and expect two solutions each period:
$\frac{180-0}{180} \times 2 = 2$ solutions.

**34 b)**

Period is $\frac{360}{3} = 120°$ and expect two solutions each period:
$\frac{360-0}{120} \times 2 = 6$ solutions.

**34 c)**

Period is $\frac{360}{4} = 90°$ and expect two solutions each period:
$\frac{720-0}{90} \times 2 = 16$ solutions.

**35 a)**

Period is $\frac{180}{2} = 90°$ and expect **one** solution each period:
$\frac{1080-0}{90} = 12$ solutions.

**35 b)**

Period is $\frac{180}{3} = 60°$ and expect **one** solution each period:
$\frac{720-0}{60} = 12$ solutions.

**35 c)**

Period is $\frac{180}{4} = 45°$ and expect **one** solution each period:
$\frac{360-(-270)}{45} = 14$ solutions.

**36 a)**

Period is 360°.

**36 b)**

Period is 360°.

**36 c)**

Period is 180°.

**36 d)**

Period is $360°$.

**36 e)**

Period is $360°$.

**36 f)**

Period is $180°$.

**36 g)**

Period is $\frac{360}{2} = 180°$.

**36 h)**

Period is $\frac{360}{4} = 90°$.

**36 i)**

Period is $\frac{180}{9} = 20°$.

**36 j)**

Period is $\frac{360}{\frac{1}{5}} = 1800°$.

**36 k)**

Period is $\frac{360}{\frac{1}{8}} = 2880°$.

**36 l)**

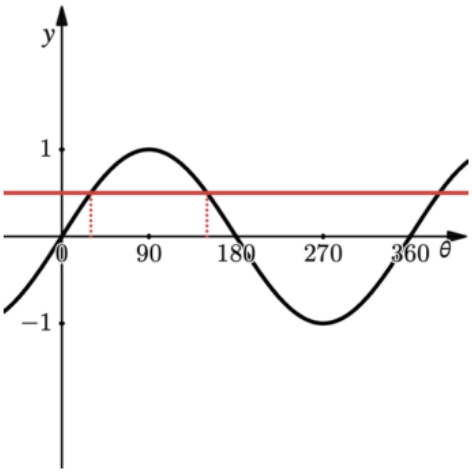
Period is $\frac{180}{\frac{5}{9}} = 324°$.

**37 a)**

$\sin(\theta + 10°) = 0.3$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$\theta + 10° = \arcsin 0.3 = 17.457\ldots°$.




Find second value between $0°$ and $360°$:

$\theta + 10° = 180° - 17.457\ldots° = 162.542\ldots°$.

Apply transformation:

$\theta + 10° = 17.457\ldots° \Rightarrow \theta = 7.457\ldots°$ (in range).

$\theta + 10° = 162.542\ldots° \Rightarrow \theta = 152.542\ldots°$ (in range).

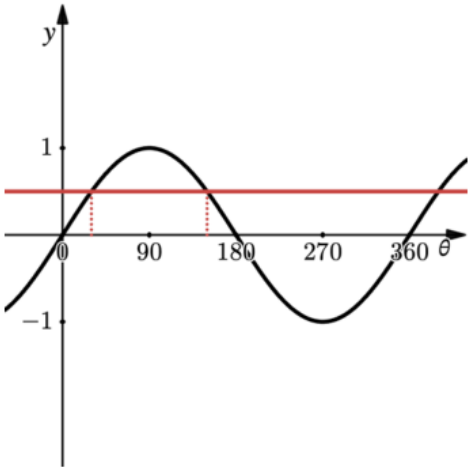
$\theta = \{7.46°, 153°\}$ (3 SF).

**37 b)**

$\sin(\theta - 50°) = \frac{2}{11}$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$\theta - 50° = \arcsin \frac{2}{11} = 10.475\ldots°$.




Find second value between $0°$ and $360°$:

$\theta - 50° = 180° - 10.475\ldots° = 169.524\ldots°$.

Apply transformation:

$\theta - 50° = 10.475\ldots° \Rightarrow \theta = 60.475\ldots°$ (in range).

$\theta - 50° = 169.524\ldots° \Rightarrow \theta = 219.524\ldots°$ (in range).

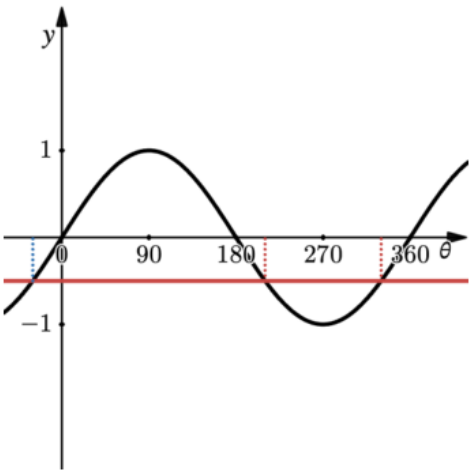
$\theta = \{60.5°, 220°\}$ (3 SF).

**37 c)**

$\sin(\theta + 205°) = -0.4$ for $0° \leq \theta < 360°$.

Find first value:

$\theta + 205° = \arcsin(-0.4) = -23.578\ldots°$.




Find first value between 0° and 360°:

$\theta + 205° = 180° + 23.578\ldots° = 203.578\ldots°$.

Find second value between 0° and 360°:

$\theta + 205° = 360° - 23.578\ldots° = 336.421\ldots°$.

Apply transformation:

$\theta + 205° = 203.578\ldots° \Rightarrow \theta = -1.421\ldots°$ (not in range).

$\theta + 205° = 336.421\ldots° \Rightarrow \theta = 131.421\ldots°$ (in range).

Add period:

$\theta = -1.421\ldots° + 360° = 358.578\ldots°$ (in range).

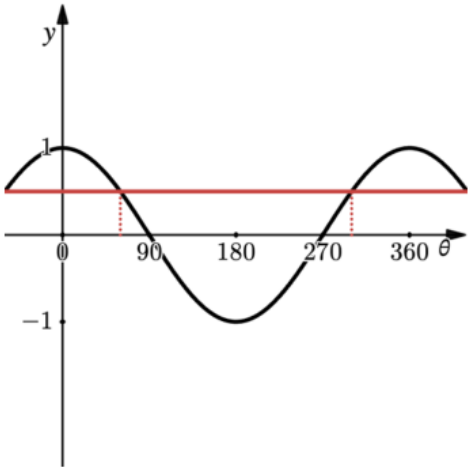
$\theta = \{131°, 359°\}$ (3 SF).

**38 a)**

$\cos(\theta - 60°) = \frac{4}{11}$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$\theta - 60° = \arccos \frac{4}{11} = 68.676\ldots°$.




Find second value between $0°$ and $360°$:

$\theta - 60° = 360° - 68.676\ldots° = 291.323\ldots°$.

Apply transformation:

$\theta - 60° = 68.676\ldots° \Rightarrow \theta = 128.676\ldots°$ (in range).

$\theta - 60° = 291.323\ldots° \Rightarrow \theta = 351.323\ldots°$ (in range).

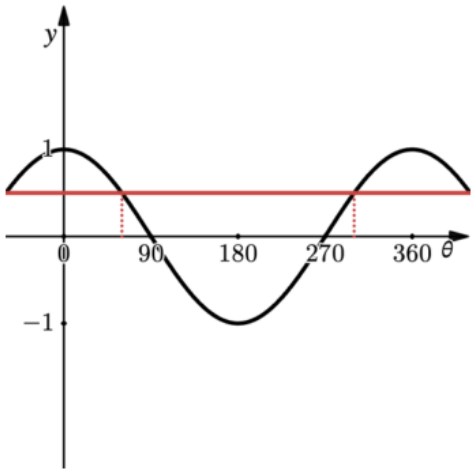
$\theta = \{129°, 351°\}$ (3 SF).

**38 b)**

$\cos(\theta + 25°) = 0.85$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$\theta + 25° = \arccos 0.85 = 31.788\ldots°$.




Find second value between $0°$ and $360°$:

$\theta + 25° = 360° - 31.788\ldots° = 328.211\ldots°$.

Apply transformation:

$\theta + 25° = 31.788\ldots° \Rightarrow \theta = 6.788\ldots°$ (in range).

$\theta + 25° = 328.211\ldots° \Rightarrow \theta = 303.211\ldots°$ (in range).

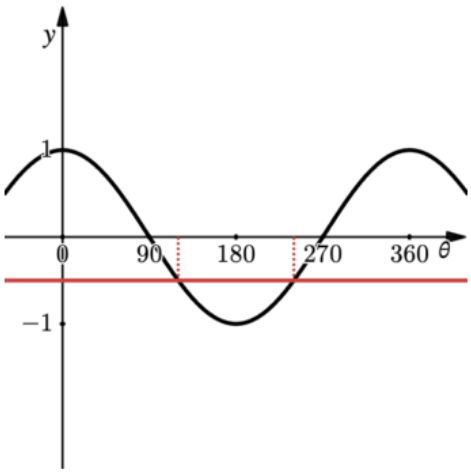
$\theta = \{6.79°, 303°\}$ (3 SF).

**38 c)**

$\cos(\theta - 195°) = -\frac{7}{12}$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$\theta - 195° = \arccos(-\frac{7}{12}) = 125.685\ldots°$.




Find second value between $0°$ and $360°$:

$\theta - 195° = 360° - 125.685\ldots° = 234.314\ldots°$.

Apply transformation:

$\theta - 195° = 125.685\ldots° \Rightarrow \theta = 320.685\ldots°$ (in range).

$\theta - 195° = 234.314\ldots° \Rightarrow \theta = 429.314\ldots°$ (not in range).

Subtract period:

$\theta = 429.314\ldots° - 360° = 69.314\ldots°$ (in range).

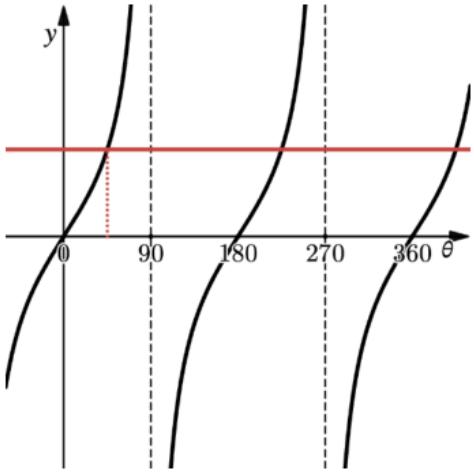
$\theta = \{69.3°, 321°\}$ (3 SF).

**39 a)**

$\tan(\theta - 45°) = 2.5$ for $0° \leq \theta < 360°$.

Find first value:

$\theta - 45° = \arctan 2.5 = 68.198\ldots°$.




Apply transformation:

$\theta - 45° = 68.198\ldots° \Rightarrow \theta = 113.198\ldots°$ (in range).

Add period:

$\theta = 113.198\ldots° + 180° = 293.198\ldots°$ (in range).

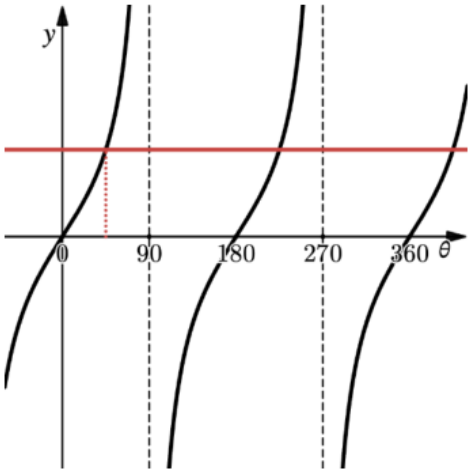
$\theta = \{113°, 293°\}$ (3 SF).

**39 b)**

$\tan(\theta + 80°) = \frac{5}{6}$ for $0° \leq \theta < 360°$.

Find first value:

$\theta + 80° = \arctan \frac{5}{6} = 39.805\ldots°$.




Apply transformation:

$\theta + 80° = 39.805\ldots° \Rightarrow \theta = -40.194\ldots°$ (not in range).

Add period:

$\theta = -40.194\ldots° + 180° = 139.805\ldots°$ (in range).

$\theta = 139.805\ldots° + 180° = 319.805\ldots°$ (in range).

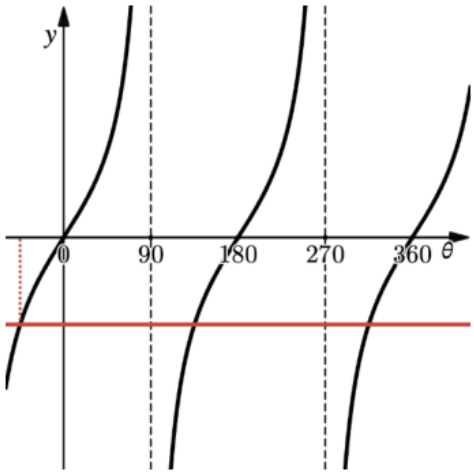
$\theta = \{140°, 320°\}$ (3 SF).

**39 c)**

$\tan(\theta - 215°) = -4.2$ for $0° \leq \theta < 360°$.

Find first value:

$\theta - 215° = \arctan(-4.2) = -76.607\ldots°$.




Apply transformation:

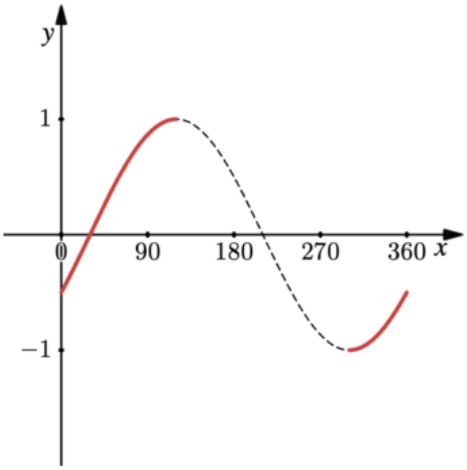
$\theta - 215° = -76.607\ldots° \Rightarrow \theta = 138.392\ldots°$ (in range).

Add period:

$\theta = 138.392\ldots° + 180° = 318.392\ldots°$ (in range).

$\theta = \{138°, 318°\}$ (3 SF).

**40)**



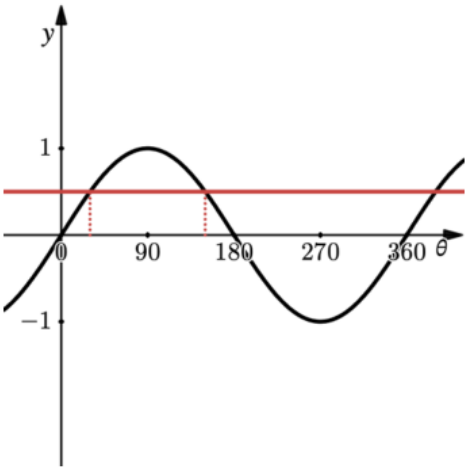

$0° \leq x < 120°$ or $300° < x \leq 360°$.

**41 a)**

$\sin 2\theta = \frac{\sqrt{2}}{2}$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$2\theta = \arcsin \frac{\sqrt{2}}{2} = 45°$.




Find second value between $0°$ and $360°$:

$2\theta = 180° - 45° = 135°$.

Apply transformation:

$2\theta = 45° \Rightarrow \theta = 22.5°$ (in range).

$2\theta = 135° \Rightarrow \theta = 67.5°$ (in range).

Add period:

$\theta = 22.5° + 180° = 202.5°$ (in range).

$\theta = 67.5° + 180° = 247.5°$ (in range).

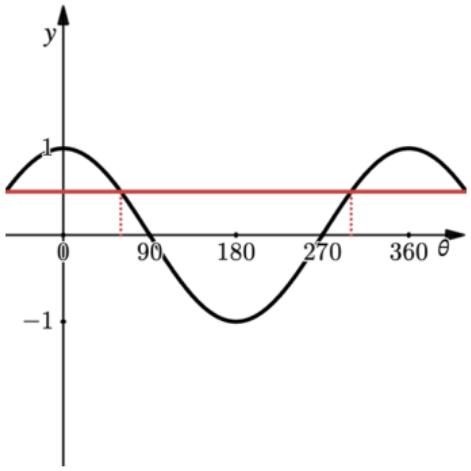
$\theta = \{22.5°, 67.5°, 202.5°, 247.5°\}$.

**41 b)**

$\cos 2\theta = \frac{\sqrt{2}}{4}$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$2\theta = \arccos \frac{\sqrt{2}}{4} = 69.295\ldots°$.




Find second value between $0°$ and $360°$:

$2\theta = 360° - 69.295\ldots° = 290.704\ldots°$.

Apply transformation:

$2\theta = 69.295\ldots° \Rightarrow \theta = 34.647\ldots°$ (in range).

$2\theta = 290.704\ldots° \Rightarrow \theta = 145.352\ldots°$ (in range).

Add period:

$34.647\ldots° + 180° = 214.647\ldots°$ (in range).

$145.352\ldots° + 180° = 325.352\ldots°$ (in range).

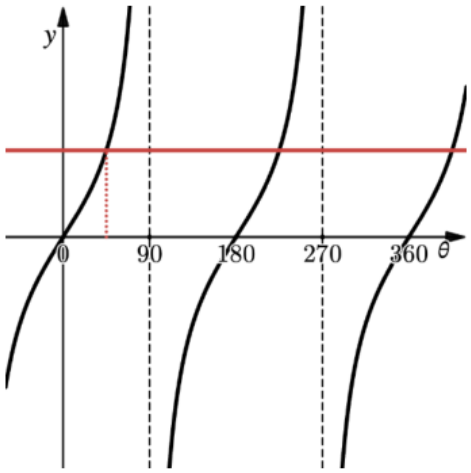
$\theta = \{34.6°, 145°, 215°, 325°\}$ (3 SF).

**41 c)**

$\tan 2\theta = \frac{\sqrt{3}}{3}$ for $0° \leq \theta < 360°$.

Find first value:

$2\theta = \arctan \frac{\sqrt{3}}{3} = 30°$.




Apply transformation:

$2\theta = 30° \Rightarrow \theta = 15°$ (in range).

Add period:

$\theta = 15° + 90° = 105°$ (in range).

$\theta = 105° + 90° = 195°$ (in range).

$\theta = 195° + 90° = 285°$ (in range).

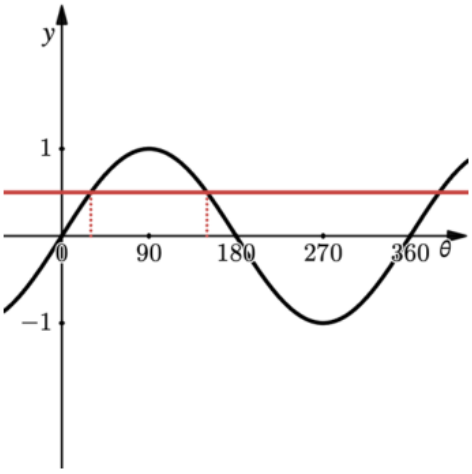
$\theta = \{15°, 105°, 195°, 285°\}$.

**42 a)**

$\sin 3\theta = 0.8$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$3\theta = \arcsin 0.8 = 53.130\ldots°$.




Find second value between $0°$ and $360°$:

$3\theta = 180° - 53.130\ldots° = 126.869\ldots°$.

Apply transformation:

$3\theta = 53.130\ldots° \Rightarrow \theta = 17.710\ldots°$ (in range).

$3\theta = 126.869° \Rightarrow \theta = 42.289\ldots°$ (in range).

Add period:

$\theta = 17.710\ldots° + 120° = 137.710\ldots°$ (in range).

$\theta = 137.710\ldots° + 120° = 257.710\ldots°$ (in range).

$\theta = 42.289\ldots° + 120° = 162.289\ldots°$ (in range).

$\theta = 162.289\ldots° + 120° = 282.289\ldots°$ (in range).

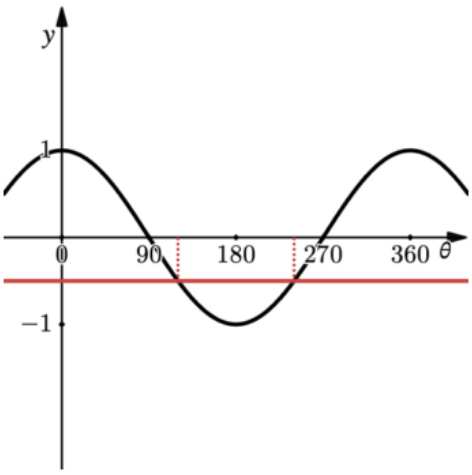
$\theta = \{17.7°, 42.3°, 138°, 162°, 258°, 282°\}$ (3 SF).

**42 b)**

$\cos 3\theta = -0.3$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$3\theta = \arccos(-0.3) = 107.457\ldots°$.




Find second value between $0°$ and $360°$:

$3\theta = 360° - 107.457\ldots° = 252.542\ldots°$.

Apply transformation:

$3\theta = 107.457\ldots° \Rightarrow \theta = 35.819\ldots°$ (in range).

$3\theta = 252.542\ldots° \Rightarrow \theta = 84.180\ldots°$ (in range).

Add period:

$\theta = 35.819\ldots° + 120° = 155.819\ldots°$ (in range).

$\theta = 84.180\ldots° + 120° = 204.180\ldots°$ (in range).

$\theta = 155.819\ldots° + 120° = 275.819\ldots°$ (in range).

$\theta = 204.180\ldots° + 120° = 324.180\ldots°$ (in range).

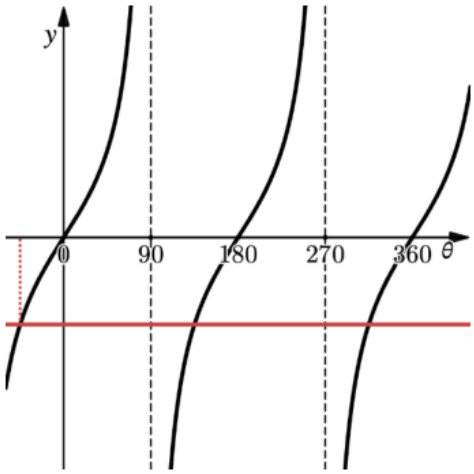
$\theta = \{35.8°, 84.2°, 156°, 204°, 276°, 324°\}$ (3 SF).

**42 c)**

$\tan 3\theta = -3$ for $0° \leq \theta < 360°$.

Find first value:

$3\theta = \arctan(-3) = -71.565\ldots°$.




Apply transformation:

$3\theta = -71.565\ldots° \Rightarrow \theta = -23.855\ldots°$ (not in range).

Add period:

$\theta = -23.855\ldots° + 60° = 36.144\ldots°$ (in range).

$\theta = 36.144\ldots° + 60° = 96.144\ldots°$ (in range).

$\theta = 96.144\ldots° + 60° = 156.144\ldots°$ (in range).

$\theta = 156.144\ldots° + 60° = 216.144\ldots°$ (in range).

$\theta = 216.144\ldots° + 60° = 276.144\ldots°$ (in range).

$\theta = 276.144\ldots° + 60° = 336.144\ldots°$ (in range).

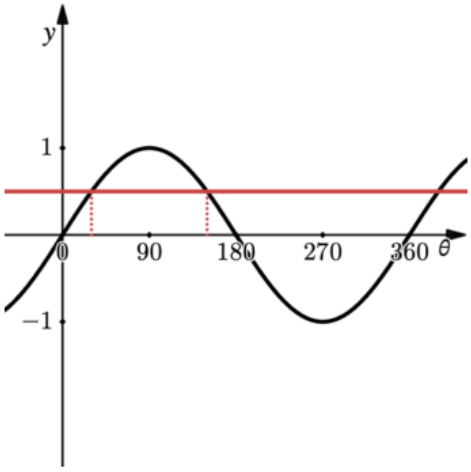
$\theta = \{36.1°, 96.1°, 156°, 216°, 276°, 336°\}$ (3 SF).

**43 a)**

$\sin \frac{1}{2}\theta = 0.2$ for $0° \leq \theta < 720°$.

Find first value between $0°$ and $360°$:

$\frac{1}{2}\theta = \arcsin 0.2 = 11.536\ldots°$.




Find second value between $0°$ and $360°$:

$\frac{1}{2}\theta = 180° - 11.536\ldots° = 168.463\ldots°$.

Apply transformation:

$\frac{1}{2}\theta = 11.536\ldots° \Rightarrow \theta = 23.073\ldots°$ (in range).

$\frac{1}{2}\theta = 168.463\ldots° \Rightarrow \theta = 336.926\ldots°$ (in range).

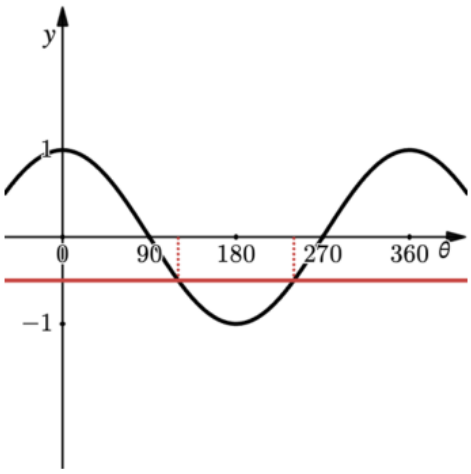
$\theta = \{23.1°, 337°\}$ (3 SF).

**43 b)**

$\cos \frac{1}{4}\theta = -\frac{4}{5}$ for $0° \leq \theta < 3600°$.

Find first value between 0° and 360°:

$\frac{1}{4}\theta = \arccos(-\frac{4}{5}) = 143.130\ldots°$.




Find second value between 0° and 360°:

$\frac{1}{4}\theta = 360° - 143.130\ldots° = 216.869\ldots°$.

Apply transformation:

$\frac{1}{4}\theta = 143.130\ldots° \Rightarrow \theta = 572.520\ldots°$ (in range).

$\frac{1}{4}\theta = 216.869\ldots° \Rightarrow \theta = 867.479\ldots°$ (in range).

Add period:

$\theta = 572.520\ldots° + 1440° = 2012.520\ldots°$ (in range).

$\theta = 2012.520\ldots° + 1440° = 3452.520\ldots°$ (in range).

$\theta = 867.479\ldots° + 1440° = 2307.479\ldots°$ (in range).

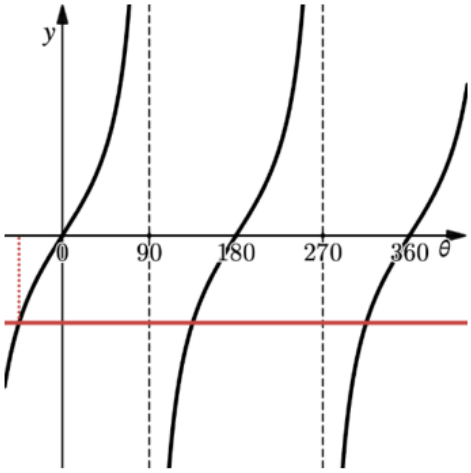
$\theta = \{573°, 867°, 2010°, 2310°, 3450°\}$ (3 SF).

**43 c)**

$\tan \frac{2}{3}\theta = -5$ for $0° \leq \theta < 270°$.

Find first value:

$\frac{2}{3}\theta = \arctan(-5) = -78.690\ldots°$.




Apply transformation:

$\frac{2}{3}\theta = -78.690\ldots° \Rightarrow \theta = -118.035\ldots°$ (not in range).

Add period:

$\theta = -118.035\ldots° + 270° = 151.964\ldots°$ (in range).

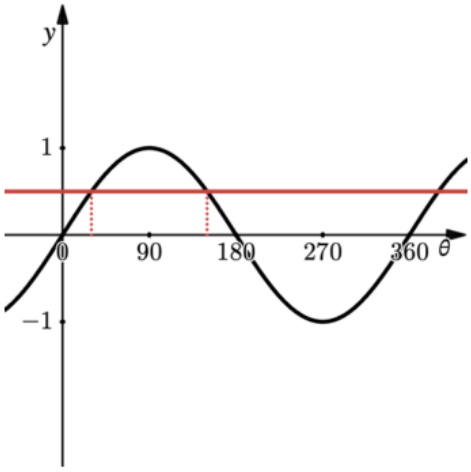
$\theta = \{152°\}$ (3 SF).

**44 a)**

$\sin(2\theta - 25°) = \frac{1}{2}$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$2\theta - 25° = \arcsin \frac{1}{2} = 30°$.




Find second value between $0°$ and $360°$:

$2\theta - 25° = 180° - 30° = 150°$.

Apply transformation:

$2\theta - 25° = 30° \Rightarrow \theta = 27.5°$ (in range).

$2\theta - 25° = 150° \Rightarrow \theta = 87.5°$ (in range).

Add period:

$\theta = 27.5° + 180° = 207.5°$ (in range).

$\theta = 87.5° + 180° = 267.5$ (in range).

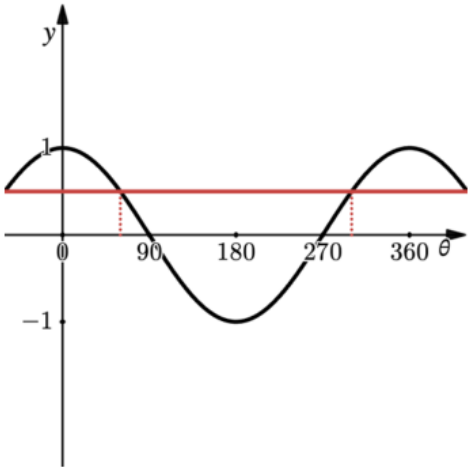
$\theta = \{27.5°, 87.5°, 207.5°, 267.5°\}$.

**44 b)**

$\cos(3\theta + 50°) = \frac{1}{2}$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$3\theta + 50° = \arccos \frac{1}{2} = 60°$.




Find second value between $0°$ and $360°$:

$3\theta + 50° = 360° - 60° = 300°$.

Apply transformation:

$3\theta + 50° = 60° \Rightarrow \theta = 3.333\ldots°$ (in range).

$3\theta + 50° = 300° \Rightarrow \theta = 83.333\ldots°$ (in range).

Add period:

$\theta = 3.333\ldots° + 120° = 123.333\ldots°$.

$\theta = 83.333\ldots° + 120° = 203.333\ldots°$.

$\theta = 123.333\ldots° + 120° = 243.333\ldots°$.

$\theta = 203.333\ldots° + 120° = 323.333\ldots°$.

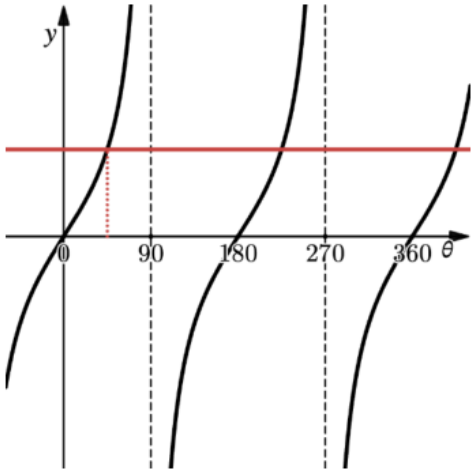
$\theta = \{3.33°, 83.3°, 123°, 203°, 243°, 323°\}$ (3 SF).

**44 c)**

$\tan(2\theta + 65°) = 1$ for $0° \leq \theta < 360°$.

Find first value:

$2\theta + 65° = \arctan 1 = 45°$.




Apply transformation:

$2\theta + 65° = 45° \Rightarrow \theta = -10°$ (not in range).

Add period:

$\theta = -10° + 90° = 80°$ (in range).

$\theta = 80° + 90° = 170°$ (in range).

$\theta = 170° + 90° = 260°$ (in range).

$\theta = 260° + 90° = 350°$ (in range).

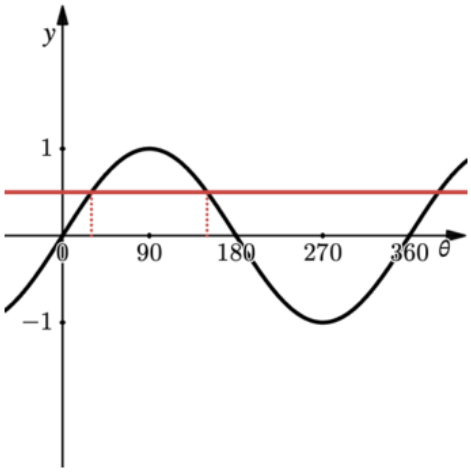
$\theta = \{80°, 170°, 260°, 350°\}$.

**45 a)**

$\sin(4\theta + 161°) = \frac{5}{6}$ for $0° \leq \theta < 180°$.

Find first value between 0° and 360°:

$4\theta + 161° = \arcsin \frac{5}{6} = 56.442\ldots°$.




Find second value between 0° and 360°:

$4\theta + 161° = 180° - 56.442\ldots° = 123.557\ldots°$.

Apply transformation:

$4\theta + 161° = 56.442\ldots° \Rightarrow \theta = -26.139\ldots°$ (not in range).

$4\theta + 161° = 123.557\ldots° \Rightarrow \theta = -9.360\ldots°$ (not in range).

Add period:

$\theta = -9.360\ldots° + 90° = 80.639\ldots°$ (in range).

$\theta = -26.139\ldots° + 90° = 63.860\ldots°$ (in range).

$\theta = 80.639\ldots° + 90° = 170.639\ldots°$ (in range).

$\theta = 63.860\ldots° + 90° = 153.860\ldots°$ (in range).

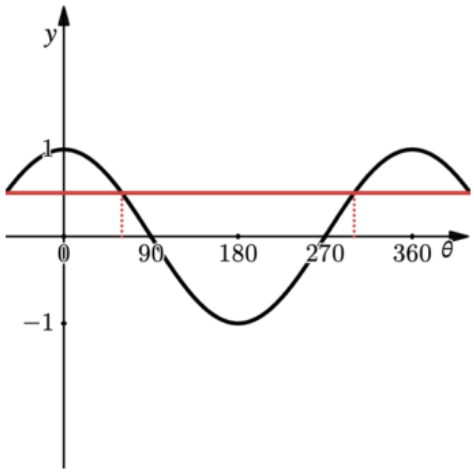
$\theta = \{63.9°, 80.6°, 154°, 171°\}$ (3 SF).

**45 b)**

$\cos(6\theta - 94°) = 0.03$ for $-90° \leq \theta < 90°$.

Find first value between $0°$ and $360°$:

$6\theta - 94° = \arccos 0.03 = 88.280\ldots°$.




Find second value between $0°$ and $360°$:

$6\theta - 94° = 360° - 88.280\ldots° = 271.719\ldots°$.

Apply transformation:

$6\theta - 94° = 88.280\ldots° \Rightarrow \theta = 30.380\ldots°$ (in range).

$6\theta - 94° = 271.719\ldots° \Rightarrow \theta = 60.953\ldots°$ (in range).

Subtract period:

$\theta = 30.380\ldots° - 60° = -29.619\ldots°$ (in range).

$\theta = -29.619\ldots° - 60° = -89.619\ldots°$ (in range).

$\theta = 60.953\ldots° - 60° = 0.953\ldots°$ (in range).

$\theta = 0.953\ldots° - 60° = -59.046\ldots°$ (in range).

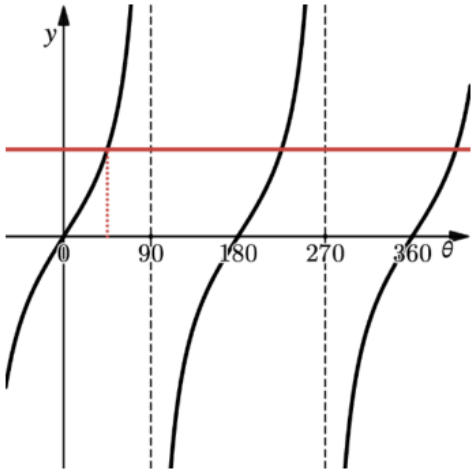
$\theta = \{-89.6°, -59.0°, -29.6°, 0.953°, 30.4°, 61.0°\}$ (3 SF).

**45 c)**

$\tan(5\theta - 213°) = 8$ for $100° \leq \theta < 200°$.

Find first value:

$5\theta - 213° = \arctan 8 = 82.874\ldots°$.




Apply transformation:

$5\theta - 213° = 82.874\ldots° \Rightarrow \theta = 59.174\ldots°$ (not in range).

Add period:

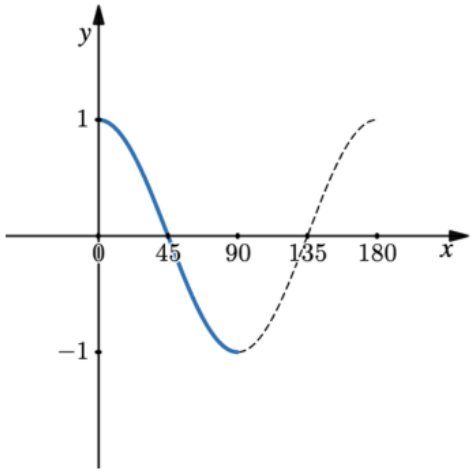
$\theta = 59.174\ldots° + 36° = 95.174\ldots°$ (not in range).

$\theta = 95.174\ldots° + 36° = 131.174\ldots°$ (in range).

$\theta = 131.174\ldots° + 36° = 167.174\ldots°$ (in range).

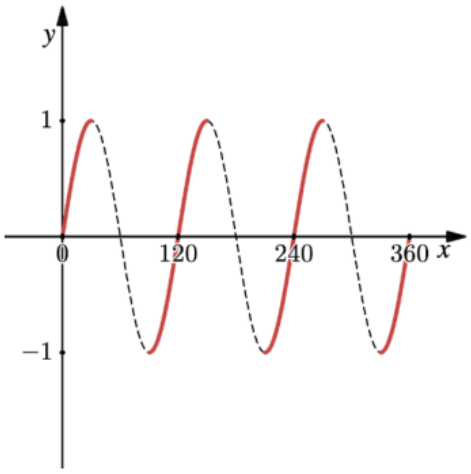
$\theta = \{131°, 167°\}$ (3 SF).

**46)**




$0° < x < 90°$.

**47)**



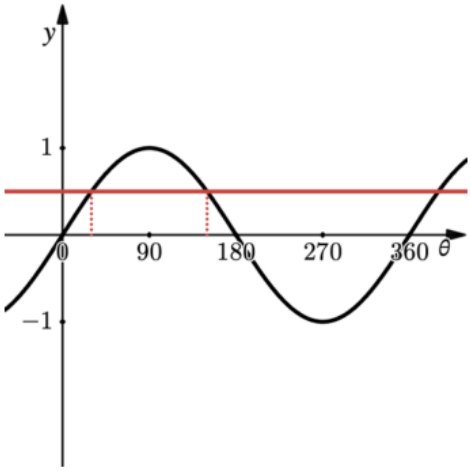

$0° < x < 30°$ or $90° < x < 150°$ or $210° < x < 270°$ or $330° < x < 360°$.

**48 a)**

$\sin(84° - 3\theta) = \frac{1}{2}$ for $0° \leq \theta < 360°$.

Find first value between 0° and 360°:

$84° - 3\theta = \arcsin \frac{1}{2} = 30°$.




Find second value between 0° and 360°:

$84° - 3\theta = 180° - 30° = 150°$.

Apply transformation:

$84° - 3\theta = 30° \Rightarrow \theta = 18°$ (in range).

$84° - 3\theta = 150° \Rightarrow \theta = -22°$ (not in range).

Add period:

$\theta = 18° + 120° = 138°$ (in range).

$\theta = -22° + 120° = 98°$ (in range).

$\theta = 138° + 120° = 258°$ (in range).

$\theta = 98° + 120° = 218°$ (in range).

$\theta = 218° + 120° = 338°$ (in range).

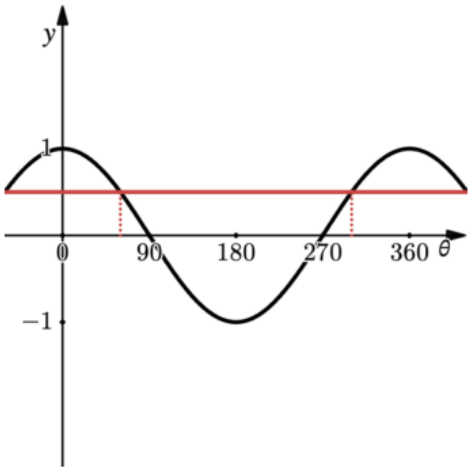
$\theta = \{18°, 98°, 138°, 218°, 258°, 338°\}$.

**48 b)**

$\cos(115° - 2\theta) = \frac{1}{2}$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$115° - 2\theta = \arccos \frac{1}{2} = 60°$.




Find second value between $0°$ and $360°$:

$115° - 2\theta = 360° - 60° = 300°$.

Apply transformation:

$115° - 2\theta = 60° \Rightarrow \theta = 27.5°$ (in range).

$115° - 2\theta = 300° \Rightarrow \theta = -92.5°$ (not in range).

Add period:

$\theta = 27.5° + 180° = 207.5°$ (in range).

$\theta = -92.5° + 180° = 87.5°$ (in range).

$\theta = 87.5° + 180° = 267.5°$ (in range).

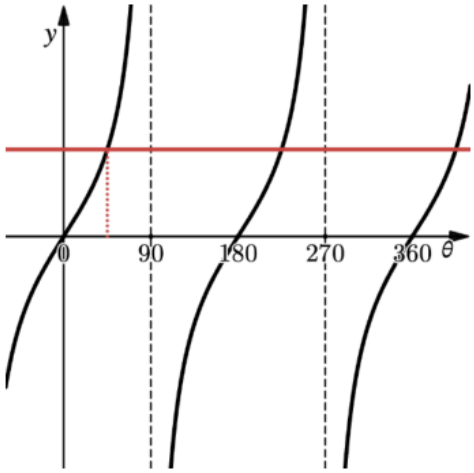
$\theta = \{27.5°, 87.5°, 207.5°, 267.5°\}$.

**48 c)**

$\tan(5° - 4\theta) = 1$ for $0° \leq \theta < 360°$.

Find first value:

$5° - 4\theta = \arctan 1 = 45°$.




Apply transformation:

$5° - 4\theta = 45° \Rightarrow \theta = -10°$ (not in range).

Add period:

$\theta = -10° + 45° = 35°$ (in range).

$\theta = 35° + 45° = 80°$ (in range).

$\theta = 80° + 45° = 125°$ (in range).

$\theta = 125° + 45° = 170°$ (in range).

$\theta = 170° + 45° = 215°$ (in range).

$\theta = 215° + 45° = 260°$ (in range).

$\theta = 260° + 45° = 305°$ (in range).

$\theta = 305° + 45° = 350°$ (in range).

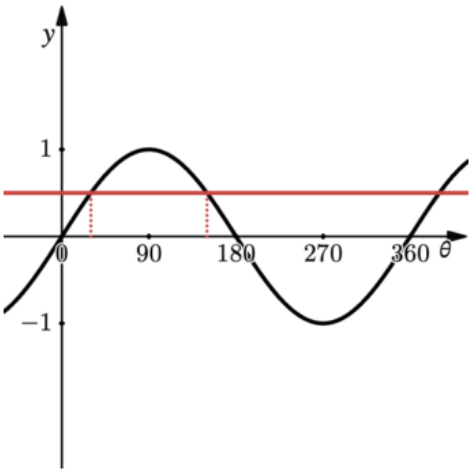
$\theta = \{35°, 80°, 125°, 170°, 215°, 260°, 305°, 350°\}$.

**49 a)**

$\sin(28° - 5\theta) = 0.1$ for $0° \leq \theta < 144°$.

Find first value between 0° and 360°:

$28° - 5\theta = \arcsin 0.1 = 5.739\ldots°$.




Find second value between 0° and 360°:

$28° - 5\theta = 180° - 5.739\ldots° = 174.260\ldots°$.

Apply transformation:

$28° - 5\theta = 5.739\ldots° \Rightarrow \theta = 4.452°$ (in range).

$28° - 5\theta = 174.260\ldots° \Rightarrow \theta = -29.252°$ (not in range).

Add period:

$\theta = 4.452\ldots° + 72° = 76.452\ldots°$ (in range).

$\theta = -29.252\ldots° + 72° = 42.747\ldots°$ (in range).

$\theta = 42.747\ldots° + 72° = 114.747\ldots°$ (in range).

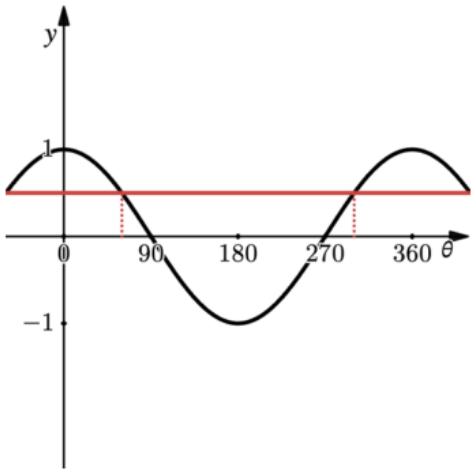
$\theta = \{4.45°, 42.7°, 76.5°, 115°\}$ (3 SF).

**49 b)**

$\cos(77° - 3\theta) = 0.99$ for $0° \leq \theta < 300°$.

Find first value between $0°$ and $360°$:

$77° - 3\theta = \arccos 0.99 = 8.109\ldots°$.




Find second value between $0°$ and $360°$:

$77° - 3\theta = 360° - 8.109\ldots° = 351.890\ldots°$.

Apply transformation:

$77° - 3\theta = 8.109\ldots° \Rightarrow \theta = 22.963\ldots°$ (in range).

$77° - 3\theta = 351.890\ldots° \Rightarrow \theta = -91.630\ldots°$ (not in range).

Add period:

$\theta = -91.630\ldots° + 120° = 28.369\ldots°$ (in range).

$\theta = 28.369\ldots° + 120° = 148.369\ldots°$ (in range).

$\theta = 148.369\ldots° + 120° = 268.369\ldots°$ (in range).

$\theta = 22.963\ldots° + 120° = 142.963\ldots°$ (in range).

$\theta = 142.963\ldots° + 120° = 262.963\ldots°$ (in range).

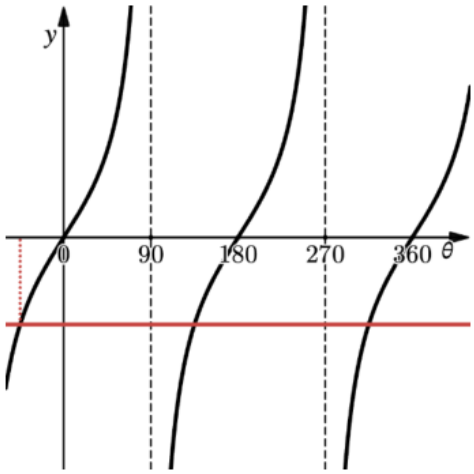
$\theta = \{23.0°, 28.3°, 143°, 148°, 263°, 268°\}$ (3 SF).

**49 c)**

$\tan(174° - 8\theta) = -0.56$ for $0° \leq \theta < 55°$.

Find first value:

$174° - 8\theta = \arctan(-0.56) = -29.248\ldots°$.




Apply transformation:

$174° - 8\theta = -29.248\ldots° \Rightarrow \theta = 25.406\ldots°$ (in range).

Add period:

$\theta = 25.406\ldots° + 22.5° = 47.906\ldots°$ (in range).

Subtract period:

$\theta = 25.406\ldots° - 22.5° = 2.906\ldots°$ (in range).

$\theta = \{2.91°, 25.4°, 47.9°\}$ (3 SF).

**50 a)**

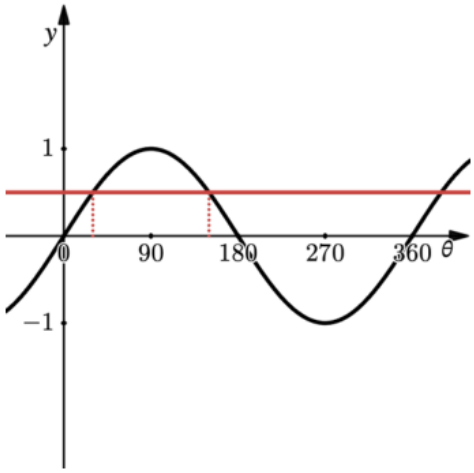
$\sin^2 2\theta - \frac{1}{4} = 0$ for $0° \leq \theta < 360°$.

$\Rightarrow \sin^2 2\theta = \frac{1}{4} \Rightarrow \sin 2\theta = \frac{1}{2}$ or $\sin 2\theta = -\frac{1}{2}$.

First solve $\sin 2\theta = \frac{1}{2}$ for $0° \leq \theta < 360°$:

Find first value between $0°$ and $360°$:

$2\theta = \arcsin \frac{1}{2} = 30°$.




Find second value between $0°$ and $360°$:

$2\theta = 180° - 30° = 150°$.

Apply transformation:

$2\theta = 30° \Rightarrow \theta = 15°$ (in range).

$2\theta = 150° \Rightarrow \theta = 75°$ (in range).

Add period:

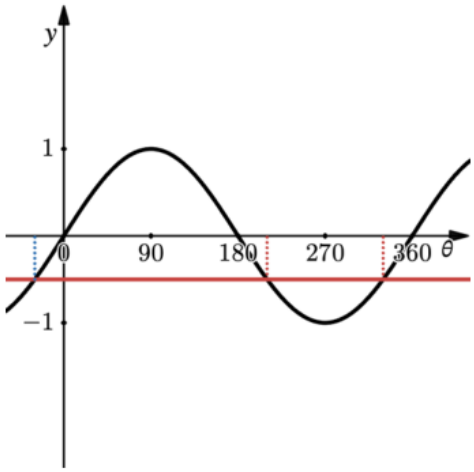
$\theta = 15° + 180° = 195°$ (in range).

$\theta = 75° + 180° = 255°$ (in range).

Now solve $\sin 2\theta = -\frac{1}{2}$ for $0° \leq \theta < 360°$:

Find first value:

$2\theta = \arcsin(-\frac{1}{2}) = -30°$.



Find first value between 0° and 360°:

$2\theta = 180° + 30° = 210°$.

Find second value between 0° and 360°:

$2\theta = 360° - 30° = 330°$.

Apply transformation:

$2\theta = 210° \Rightarrow \theta = 105°$ (in range).

$2\theta = 330° \Rightarrow \theta = 165°$ (in range).

Add period:

$\theta = 105° + 180° = 285°$ (in range).

$\theta = 165° + 180° = 345°$ (in range).

$\theta = \{15°, 75°, 105°, 165°, 195°, 255°, 285°, 345°\}$.

**50 b)**

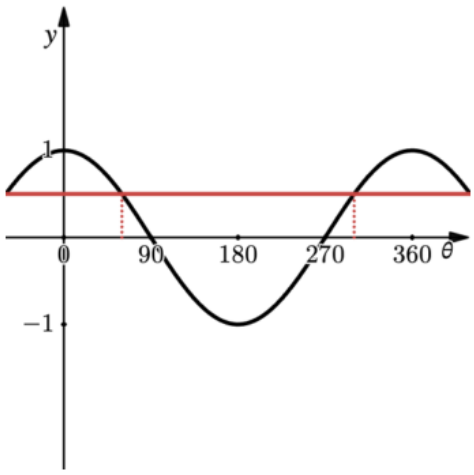
$\cos^2 3\theta - \frac{1}{16} = 0$ for $0° \leq \theta < 180°$.

$\Rightarrow \cos^2 3\theta = \frac{1}{16} \Rightarrow \cos 3\theta = \frac{1}{4}$ or $\cos 3\theta = -\frac{1}{4}$.

First solve $\cos 3\theta = \frac{1}{4}$ for $0° \leq \theta < 360°$:

Find first value between 0° and 360°:

$3\theta = \arccos \frac{1}{4} = 75.522\ldots°$.




Find second value between 0° and 360°:

$3\theta = 360° - 75.522\ldots° = 284.477\ldots°$.

Apply transformation:

$3\theta = 75.522\ldots° \Rightarrow \theta = 25.174\ldots°$ (in range).

$3\theta = 284.477\ldots° \Rightarrow \theta = 94.825\ldots°$ (in range).

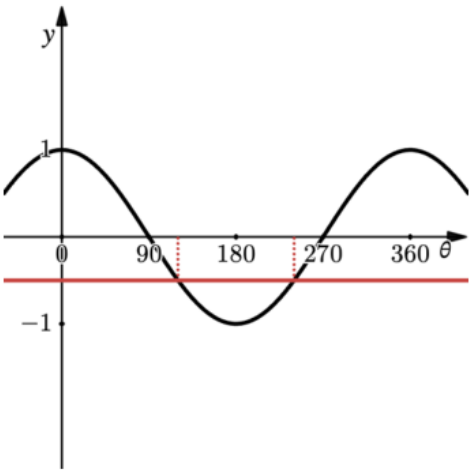
Add period:

$\theta = 25.174\ldots° + 120° = 145.174\ldots°$ (in range).

Now solve $\cos 3\theta = -\frac{1}{4}$ for $0° \leq \theta < 360°$:

Find first value between 0° and 360°:

$3\theta = \arccos(-\frac{1}{4}) = 104.477\ldots°$.




Find second value between 0° and 360°:

$3\theta = 360° - 104.477\ldots° = 255.522\ldots°$.

Apply transformation:

$3\theta = 104.477\ldots° \Rightarrow \theta = 34.825\ldots°$ (in range).

$3\theta = 255.522\ldots° \Rightarrow \theta = 85.174\ldots°$ (in range).

Add period:

$\theta = 34.825\ldots° + 120° = 154.825\ldots°$ (in range).

$\theta = \{25.2°, 34.8°, 85.2°, 94.8°, 145°, 155°\}$ (3 SF).

**50 c)**

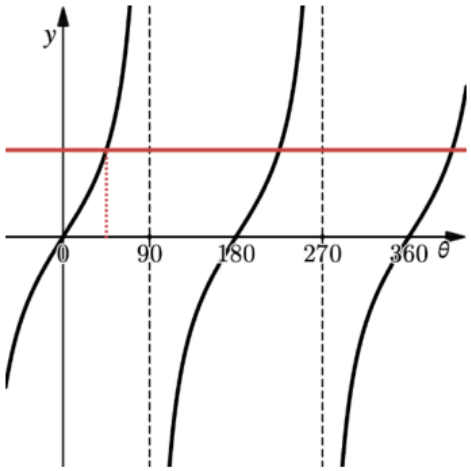
$\tan^2 5\theta - 121 = 0$ for $-40° \leq \theta < 40°$.

$\Rightarrow \tan^2 5\theta = 121 \Rightarrow \tan 5\theta = 11$ or $\tan 5\theta = -11$.

First solve $\tan 5\theta = 11$ for $-40° \leq \theta < 40°$:

Find first value:

$5\theta = \arctan 11 = 84.805\ldots°$.




Apply transformation:

$5\theta = 84.805\ldots° \Rightarrow \theta = 16.961\ldots°$ (in range).

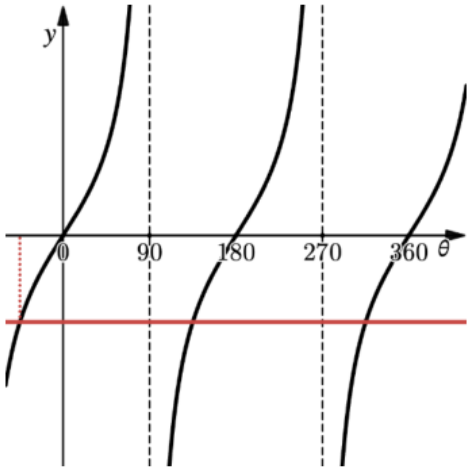
Subtract period:

$\theta = 16.961\ldots^\circ - 36^\circ = -19.038\ldots^\circ$ (in range).

Now solve $\tan 5\theta = -11$ for $-40^\circ \leq \theta < 40^\circ$:

Find first value:

$5\theta = \arctan(-11) = -84.805\ldots^\circ$.




Apply transformation:

$5\theta = -84.805\ldots^\circ \Rightarrow \theta = -16.961\ldots^\circ$ (in range).

Add period:

$\theta = -16.961\ldots^\circ + 36^\circ = 19.038\ldots^\circ$ (in range).

$\theta = \{-19.0^\circ, -17.0^\circ, 17.0^\circ, 19.0^\circ\}$ (3 SF).

**51)**

$A : x = p$.

$B : x = p - 180^\circ$.

$C : x = 90^\circ - p$.

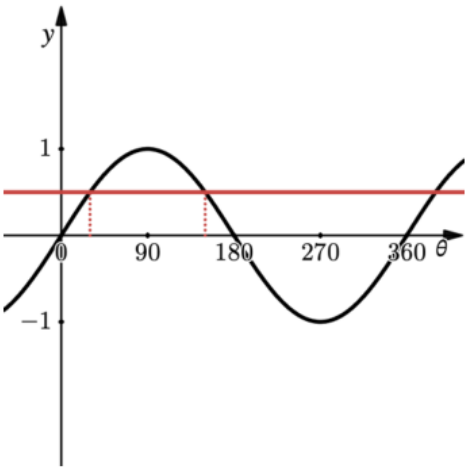
$D : x = p - 540^\circ$.

**52 a)**

$\sin(\theta - 40°) = \sin 80°$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$\theta - 40° = \arcsin(\sin 80°) = 80°$.




Find second value between $0°$ and $360°$:

$\theta - 40° = 180° - 80° = 100°$.

Apply transformation:

$\theta - 40° = 80° \Rightarrow \theta = 120°$ (in range).

$\theta - 40° = 100° \Rightarrow \theta = 140°$ (in range).

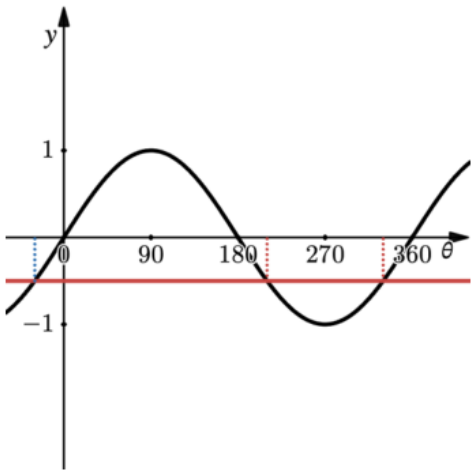
$\theta = \{120°, 140°\}$ (3 SF).

**52 b)**

$\sin(\theta + 35°) = \sin 280°$ for $0° \leq \theta < 360°$.

Find first value:

$\theta + 35° = \arcsin(\sin 280°) = -80°$.




Find first value between $0°$ and $360°$:

$\theta + 35° = 180° + 80° = 260°$.

Find second value between $0°$ and $360°$:

$\theta + 35° = 360° - 80° = 280°$.

Apply transformation:

$\theta + 35° = 260° \Rightarrow \theta = 225°$ (in range).

$\theta + 35° = 280° \Rightarrow \theta = 245°$ (in range).

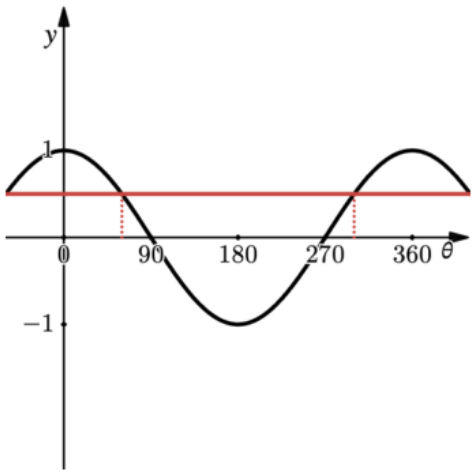
$\theta = \{225°, 245°\}$.

**52 c)**

$\cos(\theta + 15°) = \cos 50°$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$\theta + 15° = \arccos(\cos 50°) = 50°$.




Find second value between $0°$ and $360°$:

$\theta + 15° = 360° - 50° = 310°$.

Apply transformation:

$\theta + 15° = 50° \Rightarrow \theta = 35°$ (in range).

$\theta + 15° = 310° \Rightarrow \theta = 295°$ (in range).

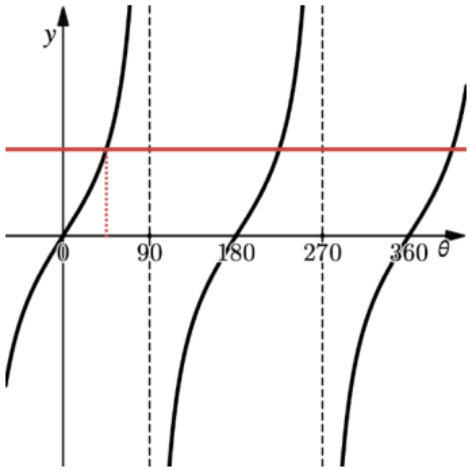
$\theta = \{35°, 295°\}$.

**52 d)**

$\tan(\theta - 70°) = \tan 20°$ for $0° \leq x < 360°$.

Find first value:

$\theta - 70° = \arctan(\tan 20°) = 20°$.




Apply transformation:

$\theta - 70° = 20° \Rightarrow \theta = 90°$ (in range).

Add period:

$\theta = 90° + 180° = 270°$ (in range).

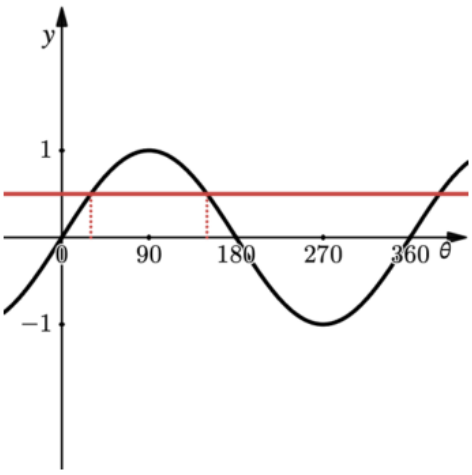
$\theta = \{90°, 270°\}$.

**53 a)**

$\sin(35° - \theta) = \sin 75°$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$35° - \theta = \arcsin(\sin 75°) = 75°$.




Find second value between $0°$ and $360°$:

$35° - \theta = 180° - 75° = 105°$.

Apply transformation:

$35° - \theta = 75° \Rightarrow \theta = -40°$ (not in range).

$35° - \theta = 105° \Rightarrow \theta = -70°$ (not in range).

Add period:

$\theta = -40° + 360° = 320°$ (in range).

$\theta = -70° + 360° = 290°$ (in range).

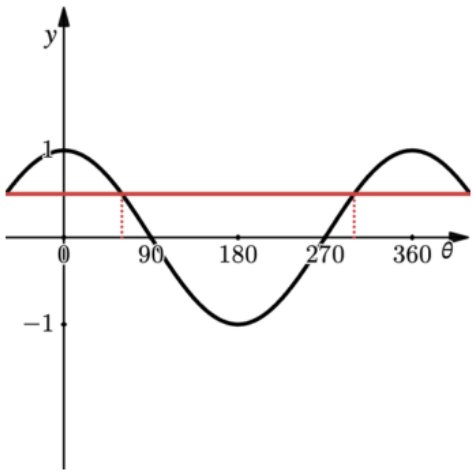
$\theta = \{290°, 320°\}$ (3 SF).

**53 b)**

$\cos(2\theta + 80°) = \cos 55°$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

$2\theta + 80° = \arccos(\cos 55°) = 55°$.




Find second value between $0°$ and $360°$:

$2\theta + 80° = 360° - 55° = 305°$.

Apply transformation:

$2\theta + 80° = 55° \Rightarrow \theta = -12.5°$ (not in range).

$2\theta + 80° = 305° \Rightarrow \theta = 112.5°$ (in range).

Add period:

$\theta = -12.5° + 180° = 167.5°$ (in range).

$\theta = 167.5° + 180° = 347.5°$ (in range).

$\theta = 112.5° + 180° = 292.5°$ (in range).

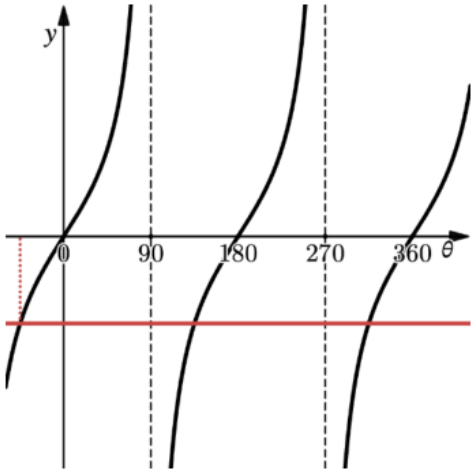
$\theta = \{112.5°, 167.5°, 292.5°, 347.5°\}$.

**53 c)**

$\tan(20° - 3\theta) = \tan 140°$ for $0° \leq \theta < 360°$.

Find first value:

$20° - 3\theta = \arctan(\tan 140°) = -40°$.




Apply transformation:

$20° - 3\theta = -40° \Rightarrow \theta = 20°$ (in range).

Add period:

$\theta = 20° + 60° = 80°$ (in range).

$\theta = 80° + 60° = 140°$ (in range).

$\theta = 140° + 60° = 200°$ (in range).

$\theta = 200° + 60° = 260°$ (in range).

$\theta = 260° + 60° = 320°$ (in range).

$\theta = \{20°, 80°, 140°, 200°, 260°, 320°\}$.

**54 a)**

$$f(\tfrac{2}{5}) = 45(\tfrac{2}{5})^3 - 63(\tfrac{2}{5})^2 - 2(\tfrac{2}{5}) + 8$$
$$= \frac{72}{25} - \frac{252}{25} - \frac{4}{5} + 8 = \frac{72 - 252 - 20 + 200}{25} = 0$$

$\therefore$ by the Factor Theorem, $(5x - 2)$ is a factor of $f(x)$.

**54 b)**

Polynomial Division:

| $\times$ | $9x^2$ | $-9x$ | $-4$ |
|---|---|---|---|
| $5x$ | $45x^3$ | $-45x^2$ | $-20x$ |
| $-2$ | $-18x^2$ | $18x$ | $8$ |

$\Rightarrow f(x) = (5x - 2)(9x^2 - 9x - 4) = (5x - 2)(3x + 1)(3x - 4)$.

$45\cos^3\theta - 63\cos^2\theta - 2\cos\theta + 8 = 0$.

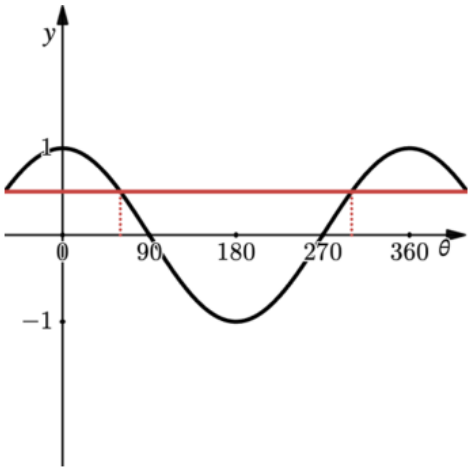
$\Rightarrow (5\cos\theta - 2)(3\cos\theta + 1)(3\cos\theta - 4) = 0.$

First solve $5\cos\theta - 2 = 0$ for $0° \leq \theta \leq 360°$:

$\Rightarrow \cos\theta = \frac{2}{5}$.

Find first value between $0°$ and $360°$:

$\theta = \arccos\frac{2}{5} = 66.421\ldots°$ (in range).




Find second value between $0°$ and $360°$:

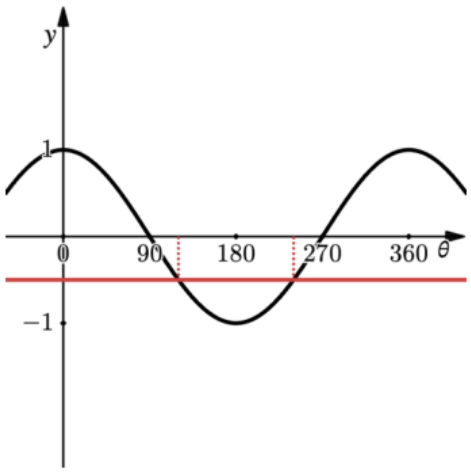
$\theta = 360° - 66.421\ldots° = 293.578\ldots°$ (in range).

Now solve $3\cos\theta + 1 = 0$ for $0° \leq \theta \leq 360°$:

$\Rightarrow \cos\theta = -\frac{1}{3}$.

Find first value between $0°$ and $360°$:

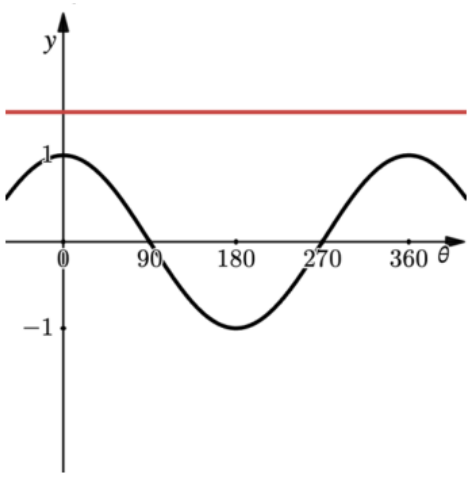
$\theta = \arccos(-\frac{1}{3}) = 109.471\ldots°$ (in range).




Find second value between $0°$ and $360°$:

$\theta = 360° - 109.471\ldots° = 250.528\ldots°$ (in range).

As for $3\cos\theta - 4 = 0 \Rightarrow \cos\theta = \frac{4}{3}$ there are no solutions.




$\Rightarrow \theta = \{66.4°, 109°, 251°, 294°\}$ (3 SF).